# BLDC MOTOR DRIVER

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**A Graduation Project Report**

**Electrical Electronics Engineering Department**

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**A Report Presented in Partial Fulfilment of the Requirements for the Degree Bachelor of Science in Electrical Electronics Engineering**

**ESKISEHIR OSMANGAZI UNIVERSITY**

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# ABSTRACT

Today, brushless direct current (BLDC) motors are used in many industrial fields, especially in the automotive sector, space technologies, the military field, and the health sector. With the rapid development in the automotive sector, the need for more efficient energy sources has emerged, leading to an increased demand for BLDC motors. The primary reasons for this trend are the efficiency, durability, and reliability of BLDC motors. However, these motors require motor drivers for effective use and control.

This project aims to gain insight into the developing BLDC motor sector and contribute to various industries, particularly the automotive and defense sectors, through the conducted studies. The main objective is to contribute to the advancement of the motor driver sector in Turkey, which is currently less developed.

Within the project activities, the structure of a general motor driver will be examined to establish the basic characteristics. Parameters such as operating frequency, output phase current, position, and speed will be considered to determine the ICs and microprocessor that provide the best efficiency. Subsequently, the circuit will be coded, and the most efficient structure will be designed through simulations and tests. Finally, the circuit will be produced, aiming to benefit for countries by controlling high-power circuits and promoting more efficient energy use.

By integrating knowledge and innovative solutions, this project aspires to be part of the advancing electric vehicle industry, enhance the local motor driver sector, and support energy efficiency and high-power circuit control.



**Keywords:** *BLDC motor, motor driver, efficiency, high-power circuit.*

# ÖZET

Günümüzde fırçasız doğru akım (BLDC) motorlar başta otomotiv sektörü, uzay teknolojileri, askeri alan ve sağlık sektörü olmak üzere birçok endüstriyel alanda kullanılmaktadır. Otomotiv sektörünün hızla gelişmesiyle birlikte daha verimli enerji kaynaklarına olan ihtiyaç ortaya çıkmış ve bu da BLDC motorlara olan talebin artmasına neden olmuştur. Bu eğilimin başlıca nedenleri BLDC motorların verimliliği, dayanıklılığı ve güvenilirliğidir. Ancak bu motorların etkin kullanımı ve kontrolü için motor sürücülerine ihtiyaç vardır.

Bu proje, gelişen BLDC motor sektörüne ışık tutmayı ve yapılan çalışmalarla başta otomotiv ve savunma sektörü olmak üzere çeşitli sektörlere katkı sağlamayı amaçlıyor. Temel amaç, Türkiye'de henüz az gelişmiş olan motor sürücü sektörünün gelişmesine katkıda bulunmaktır.

Proje faaliyetleri kapsamında genel bir motor sürücüsünün yapısı incelenerek temel özellikleri belirlenecektir. En iyi verimliliği sağlayan entegrelerin ve mikroişlemcinin belirlenmesinde çalışma frekansı, çıkış faz akımı, konum ve hız gibi parametreler dikkate alınacaktır. Daha sonra devre kodlanacak, simülasyonlar ve testler yoluyla en verimli yapı tasarlanacaktır. Son olarak yüksek güçlü devrelerin kontrol altına alınması ve daha verimli enerji kullanımının teşvik edilerek ülkemize fayda sağlanması hedeflenen devre üretilecek.

Bu proje, bilgi birikimini ve yenilikçi çözümleri entegre ederek, gelişen elektrikli araç endüstrisinin bir parçası olmayı, yerel motor sürücüsü sektörünü geliştirmeyi ve enerji verimliliğini ve yüksek güçlü devre kontrolünü desteklemeyi amaçlamaktadır.

**Anahtar Kelimeler:** *BLDC motor, motor sürücüsü, verimlilik, yüksek güç devresi*.

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# LIST OF SYMBOLS AND ABBREVIATIONS

| Symbol | Explanation |
|---|---|
| π: | pi number. |
| Ω: | Ohm |
| V: | Voltage |
| R: | Resistance |
| uH: | Microhenry |
| Rpm: | Revolutions per minute |
| Hz: | Hertz |
| F: | Farad |
| uF: | Microfarad |
| nF: | nano Farad |
| P: | Power |
| f: | Frequency |
| C: | Capacitance |
| L: | Inductance |
| η: | Efficiency |
| Ton: | On-Time |
| Toff: | Off-Time |
| mm: | Millimeter |
| cm: | Centimeter |

| **Abbreviation** | **Explanation** |
|---|---|
| DC: | Direct Current |
| AC: | Alternating Current |
| PCB: | Printed Circuit Board |
| IEEE: | Institute of Electrical and Electronics Engine |
| BLDC: | Brushless Direct Current |
| PWM: | Pulse Width Modulation |
| MOSFET: | Metal Oxide Semi-Conductor Field Effect Transistor |
| IC: | Integrated Circuit |
| MCU: | Micro Controller Unit |
| EMI: | Electromagnetic Interference |
| EMC: | Electromagnetic Compatibility |
| GPIO: | General Purpose Input/Output |
| TIM: | Timer |

# 1. INTRODUCTION

The rapid advancement of the social economy has underscored the critical importance of green sustainable development, necessitating accelerated efforts to develop and utilize green energy sources. In the field of electric vehicle (EV) technologies, efficient energy planning and the deployment of energy-efficient electric motors are crucial. Enhancing motor efficiency and minimizing losses yield two significant benefits: extending the vehicle's travel distance and reducing energy consumption per kilometer. Consequently, EV manufacturers have concentrated their research and development efforts on these objectives [1].

A promising solution in this context is the implementation of DC brushless motors, which provide numerous benefits including compact size, long lifespan, high efficiency, safety, reliability, and low noise levels. Unlike traditional motors, DC brushless motors utilize electronic commutation rather than mechanical commutation, which reduces noise and enhances efficiency [2]. These motors, powered by electric energy, are environmentally friendly and can be fully utilized without causing harm to nature. Therefore, DC brushless motors have become widely used in household appliances, automobiles, and various industrial applications.

As the need to combat environmental pollution and decrease fossil fuel consumption grows, developed countries are increasingly adopting alternative, eco-friendly transportation solutions. This trend has led to a surge in the use of electric, light electric, and hybrid vehicles on the roads, pushing automotive manufacturers to ramp up their research and development activities in this area [3].

In the application of brushless DC motors, both motor design and control are critical components. While basic power electronic circuits manage the control, there is a rising demand for advanced control algorithms [4]. The progress in microprocessor technology has enabled the application of complex mathematical models and the development of sophisticated digital controllers. These technological advancements have facilitated the creation of both cost-

effective and advanced solutions, aligning with industry demands and promoting the advancement of green and sustainable transportation technologies.

Manufacturers of power electronics components and microprocessors/microcontrollers have also responded to the growing brushless DC motor market by developing and offering specialized components for their control. These include elements tailored for the basic needs and applications of brushless DC motors such as speed control, switching element triggering, and position sensing [5]. Modern controllers, particularly microprocessors, are notable for their extensive programming capabilities, data handling capacity, and ability to integrate different control algorithms through parametric settings.

The focus of this thesis is to design and produce an efficient circuit board for the position control of brushless DC motors. Specifically, a Sensored BLDC Motor Controller has been developed in this study to provide a contemporary solution to industrial needs based on existing literature and previous research on BLDC motors. The engineering design project addresses a complex problem in electrical and electronics engineering, involving multiple disciplines such as power electronics, control systems, and digital signal processing. The integration of advanced control algorithms with real-time processing capabilities demands a deep understanding of both hardware and software aspects of motor control systems. The project involves designing a circuit that not only meets the functional requirements but also optimizes performance in terms of efficiency, reliability, and cost-effectiveness. Existing solutions for BLDC motor control typically involve basic power electronic circuits and standard microcontrollers or microprocessors. These solutions often focus on fundamental control tasks such as speed regulation and position sensing [6]. However, they may lack the sophistication required for more advanced applications, particularly in terms of integrating complex control algorithms and achieving higher levels of efficiency and precision. The proposed design shares similarities with existing solutions in its use of basic components such as microprocessors and power electronics circuits. However, it differentiates itself by integrating advanced control algorithms that enhance performance and efficiency. Unlike conventional designs, the Sensored BLDC Motor Controller developed in this project leverages the latest advancements in microprocessor technology to provide greater programming flexibility, data handling capacity, and the ability to harmonize

different control algorithms through parametric settings. This approach not only meets but exceeds the current industrial standards, offering a more robust and adaptable solution for modern electric vehicles and other applications [7].

In summary, this project aims to bridge the gap between traditional motor control systems and the next generation of high-efficiency, environmentally friendly electric drive solutions. By addressing the complex challenges of BLDC motor control, it contributes to the ongoing efforts towards sustainable development and green energy utilization.

## 2. REQUIREMENTS SPECIFICATION

A promising development in advanced motor technologies is the adoption of DC brushless motors. These motors offer advantages such as compact size, long life, high efficiency, improved safety, reliability and low noise. Unlike traditional motors, they use electronics instead of mechanical commutation, reducing noise and increasing efficiency. Products powered by electricity are environmentally friendly and do not harm nature. The integration of electronic systems and various digital designs increasingly affects many mechanical fields in daily lives, facilitating their digital transformation and offering advanced electronic functions to users. This project focuses on the development of a special control card designed specifically for BLDC electric motors, unlike the mechanical fossil fuel motors used today. This control card is capable of making real-time, user-based speed adjustments, providing a modern solution for precise and efficient motor control.

- **Physical Requirements**

The total weight of the system is around 800 gr.

The area covered by the system is designed to be approximately $12 * 25\ \text{cm} = 300\ \text{cm}\text{^}2$.

- **Performance and Functionality Requirements**

The system must read the information it receives from the hall effect sensors correctly. The over current protection must work correctly.

There must be soft start protection against sudden gas changes and starting current.

It is 24V regulator powered, resistant to 84V switching supply voltage and 40A phase current.

The temperature of the MOSFETs must be controlled.

- **Economic Requirements**

The total parts and manufacturing cost cannot exceed $300 per unit.

The total labour cost should be about $50.

- **Environmental Requirements**

This system is 10% more efficient than equivalent systems.

It can operate between -25 and 70 degrees.

It is not water resistant and cannot work underwater.

- **Health and Safety Requirements**

Since the system operates at high voltage and amperage, it is recommended not to touch it, but isolation is provided in the system.

The system is sensitive to electricity generated in the body. It is recommended to ground the body before touching.

It is not safe to operate outside the recommended temperature.

- **Manufacturability and Maintainability Requirements**

The system will be produced on a circuit board.

The integrated circuits are at a level that can be found easily.

In case of any system failure, the integrated circuits can be easily replaced.

## 3. STANDARDS

The following standards have been determined according to the literature review and studies conducted.

- **IEC 60950-1** ensures that the motor driver PCB meets the safety requirements for information technology equipment. (Insulation test was performed with a multimeter after circuit design)

- **IPC-9252** (guarantees that the PCB has passed electrical testing to ensure proper functionality). The components were tested one by one while placing them, and then unit by unit after placement, and then as a whole.

- **IEC 60309** (Dimensional interchangeability of accessories with contact rods and contact sockets) The hall effect inputs are designed as replaceable plugs.

- **TS EN 61029-1** (General rules for portable tools powered by an electric motor) The system complies with the general rules of use in portable devices.

- **IPC-6012** ensures that the motor driver PCB meets qualification and performance specifications, ensuring reliability in operation.

## 4. PATENTS

According to the literature review and studies carried out, help was taken from the following patents.

- Hall sensors that detect the rotor position of the brushless DC motor, Patent No: JP6706938B2

- Brushless DC electric motor driving control method, Patent No: KR101936476B1
- BLDC Motor Driver for Protecting From Over Voltage, Patent No: KR101269615B1

# 5. THEORETICAL BACKGROUND

### Calculation of Power Losses

The thermal loss on the mosfet is equal to the power dissipation of the mosfet. And since we put the MOSFET as the shunt resistor for current protection in the system, there are two main losses in the system: the loss on the MOSFET and the loss on the motor.

## 5.1 Power Losses of Mosfet

The power loss on the mosfet is based on 3 losses

$$P_{mos} = P_{conduction} + P_{gate} + P_{swiching} \tag{5.1.1}$$

$P_{conduction}$ is the biggest loss during conduction and is calculated as:

$$P_{conduction} = {I_d}^2 * R_{ds} \tag{5.1.2}$$

$P_{swiching}$ is calculated as follows:

$$P_{swiching} = \frac{1}{2} * V_{ds} * I_d * F_{sw} * \left(t_{rise} + t_{fall}\right) \tag{5.1.3}$$

$P_{gate}$ is calculated as follows:

$$P_{swiching} = V_{gs} * F_{sw} * Q_g \tag{5.1.4}$$

## 5.2 Thermal Losses of Mosfet

thermal losses indicate how much the mosfet will be heated:

$$P_{thermal} = P_{mos} \tag{5.2.1}$$

The maximum temperature is:

$$℃ = P_{mos} * T_{junction} \tag{5.2.2}$$

### 5.3 Thermal Losses of Regulators

The loss and efficiency of regulators is given below:

$$P_{in} = V_{in} * I_{in} \tag{5.3.1}$$

$$P_{out} = V_{out} * I_{out} \tag{5.3.2}$$

$$P_{loss} = P_{in} * P_{out} \tag{5.3.3}$$

$$\text{efficiency} = \frac{P_{out}}{P_{in}} * 100 \tag{5.3.4}$$

### 5.4 R-C snubber calculation

Snubber calculation is basically based on 7 steps:

**Step 1:** measure the circuits frequency with load($f_0$).

**Step 2:** Add a capacitor ($C_1$) in parallel with the rectifier or FET and measure the shifted oscillation frequency ($f_1$).

**Step 3:** calculate the frequency shift ratio:

$$m = \frac{f_0}{f_1} \tag{5.4.1}$$

**Step 4:** Calculate the circuit's parasitic capacitance:

$$C_0 = \frac{C_1}{(m^2 - 1)} \tag{5.4.2}$$

**Step 5:** Calculate the circuit's parasitic inductance:

$$L = \frac{(m^2 - 1)}{(2 * \pi * f_0)^2 * C_1} \tag{5.4.3}$$

**Step 6:** Calculate the starting snubber capacitor value:

$$C_{snub} = 3 * C_0 \tag{5.4.4}$$

**Step 7:** Calculate the starting snubber-resistor value:

$$C_{snub} = \sqrt{\frac{L}{C_0}} \tag{5.4.5}$$

# 6. METHODOLOGY

Emphasizing green sustainable development integrates, cutting-edge microprocessor technology with advanced control algorithms to enhance the performance, efficiency, and reliability of brushless DC motors. The systematic process involves interdisciplinary collaboration, leveraging insights from power electronics, control systems, and digital signal processing. Through a detailed exploration of both hardware and software aspects, this methodology aims to bridge the gap between traditional motor control systems and next-generation, eco-friendly electric drive solutions.

To address the critical issue of excess current and thermal losses that reduce motor efficiency, the system is designed in a sustainable and efficient manner. The result is a motor driver design with an 84V operating voltage to maximize efficiency. Motor drive design consists of two main parts:

Microprocessor section: In this section, control operations in the circuit are performed. Mosfets are opened in the appropriate order and the data coming from the motor and the users are evaluated.

Power section: This section generally consists of DC bus, MOSFETs and gate drivers. There are high voltages and amperes in this section. This should be taken into consideration when designing.

## 6.1 System Hardware

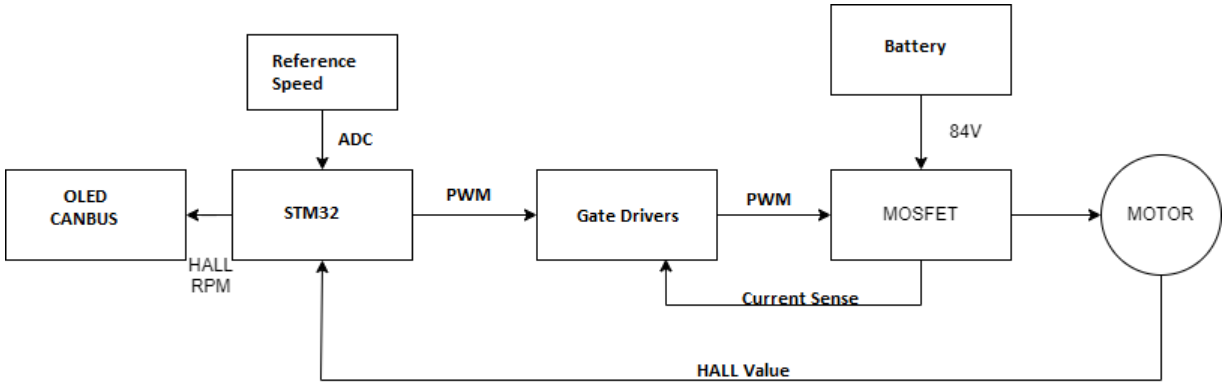



*Figure 1. Motor Driver Block Diagram*

Optimum accuracy of all these values was taken into consideration when designing the driver. The motor rotates according to the reference speed value received by the user and the Hall sensor values received by the motor. According to these values, appropriate signals are sent to the gate drivers. mosfets are driven by increasing the wavelength of these signals. In case of possible high amperage, the PWM signal is cut off at that value by gate drivers. Block diagram is shown in Figure 1

### 6.1.1 Switching Element

MOSFETs were selected as the switching elements because of their efficiency and dependability in managing the high-speed switching necessary for motor control, considering the characteristics of the BLDC motor utilized in the vehicle. Determining the proper switching frequency is crucial when employing MOSFETs in this application since it has a big impact on the motor's efficiency and performance.

The electrical characteristics of the motor, in particular the resistance (R) and inductance (L) ratio, which determine the maximum frequency the motor can handle, and the power stage

considerations, where the objective is to minimize switching losses and mitigate the effects of dead time, must be carefully considered when choosing the ideal switching frequency.

Switching frequencies higher than 10kHz are often preferred in motor control applications. Frequencies in the range of 15kHz are particularly advantageous because they are above the audible range, thereby eliminating noise that can be detected by the human ear. Operating at frequencies lower than this threshold can result in audible noise, which is undesirable in most applications. However, selecting a frequency that is too high can introduce significant challenges, such as increased switching losses and higher heat generation within the power stage components.

The choice of an optimal switching frequency involves a trade-off between reducing noise and minimizing thermal and switching losses. By selecting a frequency around 15kHz, it is possible to achieve a balance that ensures smooth and quiet operation while maintaining high efficiency and effective thermal management. This careful calibration helps in maximizing the overall performance and longevity of the BLDC motor and the associated power electronics, aligning with the goals of sustainable and efficient electric vehicle design.

In the selection process of the MOSFET, consideration was given to the potential issue of overheating. As such, a MOSFET capable of withstanding high temperatures, delivering maximum efficiency in energy transmission, and enduring high voltage was selected. It was observed that heating is influenced by the RDS(on) resistance and gate capacitance, with an increase in these parameters leading to greater heating issues. Therefore, the IRFB4110 MOSFET from INCHANGE Semiconductor was determined to be appropriate for this application. The specifications of the MOSFET features are shown in Figure 2, Table 1 and 2.

- Drain Current –$I_D$= 180A@ $T_C$=25℃
- Drain Source Voltage-
  : $V_{DSS}$= 100V(Min)
- Static Drain-Source On-Resistance
  : $R_{DS(on)}$ = 4.5mΩ (Max)

*Figure 2. IRFB4110 Mosfet Features*

*Table 1. IRFB4110 Mosfet voltage/current retying values*

**ABSOLUTE MAXIMUM RATINGS($T_a$=25℃)**

| SYMBOL | PARAMETER | VALUE | UNIT |
|---|---|---|---|
| $V_{DSS}$ | Drain-Source Voltage | 100 | V |
| $V_{GS}$ | Gate-Source Voltage-Continuous | ±20 | V |
| $I_D$ | Drain Current-Continuous | 180 | A |
| $I_{DM}$ | Drain Current-Single Pluse | 670 | A |
| $P_D$ | Total Dissipation @$T_C$=25℃ | 370 | W |
| $T_J$ | Max. Operating Junction Temperature | 175 | ℃ |
| $T_{stg}$ | Storage Temperature | -55~175 | ℃ |

*Table 2. IRFB4110 Mosfet Electrical Characteristics*

**ELECTRICAL CHARACTERISTICS**

**$T_C$=25℃ unless otherwise specified**

| SYMBOL | PARAMETER | CONDITIONS | MIN | TYPE | MAX | UNIT |
|---|---|---|---|---|---|---|
| $V_{(BR)DSS}$ | Drain-Source Breakdown Voltage | $V_{GS}$= 0; $I_D$= 0.25mA | 100 | | | V |
| $V_{GS(th)}$ | Gate Threshold Voltage | $V_{DS}$= $V_{GS}$; $I_D$= 0.25mA | 2 | | 4 | V |
| $R_{DS(on)}$ | Drain-Source On-Resistance | $V_{GS}$= 10V; $I_D$= 75A | | | 4.5 | mΩ |
| $I_{GSS}$ | Gate-Body Leakage Current | $V_{GS}$= ±20V;$V_{DS}$= 0 | | | ±100 | nA |
| $I_{DSS}$ | Zero Gate Voltage Drain Current | $V_{DS}$= 100V; $V_{GS}$= 0 | | | 20 | μA |
| $V_{SD}$ | Forward On-Voltage | $I_S$= 75A; $V_{GS}$=0 | | | 1.3 | V |
| Gfs | Forward Transconductance | $V_{DS}$= 50V； $I_D$= 75A | 160 | | | S |

### 6.1.2 Switching Gate Driver

To ensure stable operation of the MOSFETs, a higher voltage must be applied to the gate pin relative to the source pin. To achieve this, a gate driver with a high-side driver capability, specifically the IR2127, was selected. This gate driver is capable of providing the necessary voltage to the MOSFET's gate pin above the source pin. The IR2127 gate driver operates with a supply voltage ranging from 12 to 20 volts and includes built-in current sensing capabilities. Its turn-on (Ton) and turn-off (Toff) times are 200 ns and 150 ns, respectively. These characteristics, along with its ease of use and integrated functionalities, make the IR2127 gate driver a suitable choice for this application. The features of the gate driver are given below Figure 3 and 4.

| | | |
|---|---|---|
| $V_{OFFSET}$ | 600V max. | |
| $I_O$+/- | 200 mA / 420 mA | |
| $V_{OUT}$ | 12 - 20V (IR2127/IR2128) | 9 - 20V (IR21271) |
| $V_{CSth}$ | 250 mV or 1.8V | |
| $t_{on/off}$ (typ.) | 200 & 150 ns | |

*Figure 3. IR2127 Features 1*

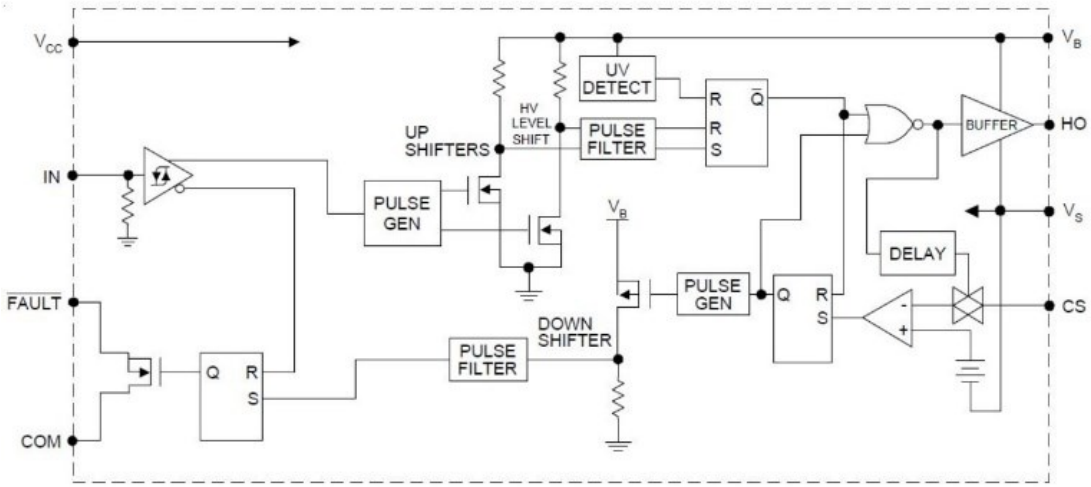



*Figure 4. IR2127 Features 2*

### 6.1.2 Switching Circuit

Since ensuring MOSFET safety is a crucial component of driver design, the MOSFETs were driven using a gate driver combined with bootstrap technology. The objectives of this research are to reduce the possibility of short circuits caused by EMI/EMC problems, protect phases from reversible current, and stop interference effects. The designed circuit is given below in Figure 5.

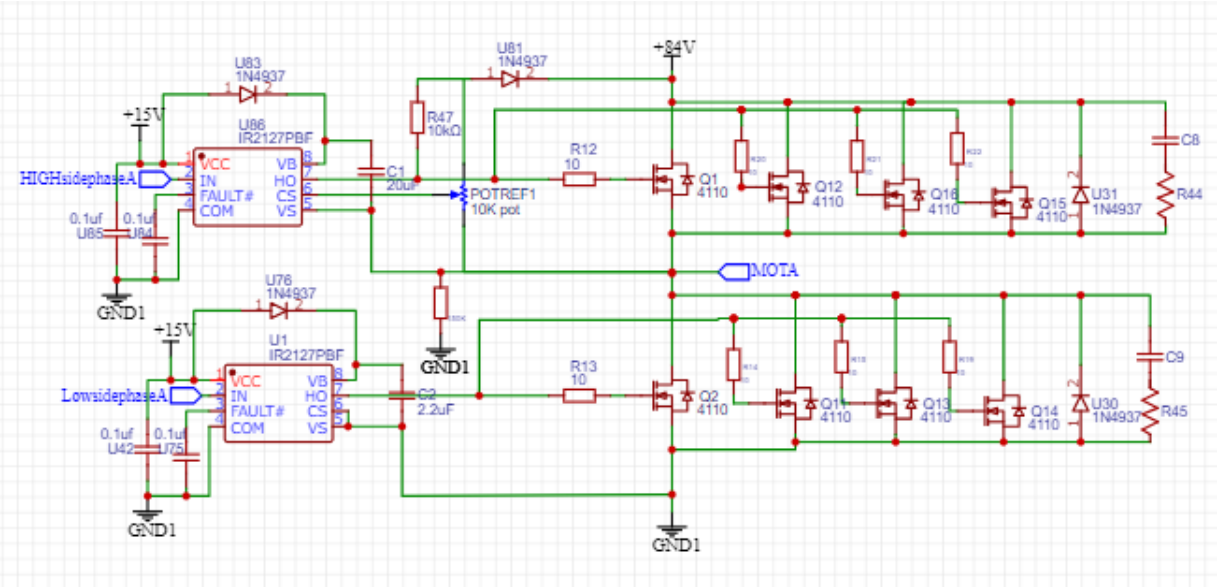



*Figure 5. Switching Circuit*

In the design of the circuit, the choice of the IR2127 is paramount, primarily due to its integrated current sensing capabilities. By incorporating this component, the circuit gains the ability to monitor and regulate current flow, ensuring optimal performance and safety.

The decision to utilize four MOSFETs in parallel within the circuit architecture is a strategic one, aimed at minimizing power dissipation and enhancing current handling capacity. By distributing the load across multiple MOSFETs, the circuit achieves greater efficiency and reliability, particularly under high load conditions. To drive the MOSFETs effectively and prevent sudden current ripples, a 10 ohm resistor is introduced. This resistor acts as a buffer,

ensuring smooth and controlled operation of the MOSFETs, thereby reducing the risk of voltage spikes and other potential issues. To activate the bootstrap capacitor during the initial stages of operation, a 150k resistor parallel to the phase is employed. This ensures that the capacitor is charged and ready to provide the necessary voltage boost for proper MOSFET gate driving, enhancing overall efficiency and performance.

Furthermore, the inclusion of RC snubbers serves a crucial role in protecting the MOSFETs against sudden voltage ripples that may occur during switching transitions (Tone and Toff states). By damping these voltage spikes, the snubbers help to minimize stress on the MOSFETs and prolong their operational lifespan, thereby enhancing the reliability and robustness of the circuit. the utilization of the IR2127, coupled with the strategic incorporation of parallel MOSFETs, resistors, bootstrap capacitors, and RC snubbers, represents a comprehensive approach to circuit design aimed at optimizing performance, efficiency, and reliability.

### 6.1.3 DC-Bara

In pursuit of stable and efficient circuit operation, a DC bus system has been integrated to balance power distribution among phases and reduce voltage ripples. This system incorporates four 470µF capacitors and three 47µF capacitors connected in parallel on the busbar, chosen based on practical applications to optimize power delivery and ripple suppression. Complementing this setup, three 1MΩ resistors are strategically placed to discharge stored voltage safely, preventing overvoltage issues. Additionally, a 58nF ceramic capacitor serves as a low-pass filter, effectively suppressing high-frequency noise. Through the careful selection of component values and the implementation of effective voltage discharge and filtering mechanisms, the circuit achieves optimal stability, performance, and reliability, addressing critical challenges in power management and ripple mitigation The designed circuit is given below in Figure 6.

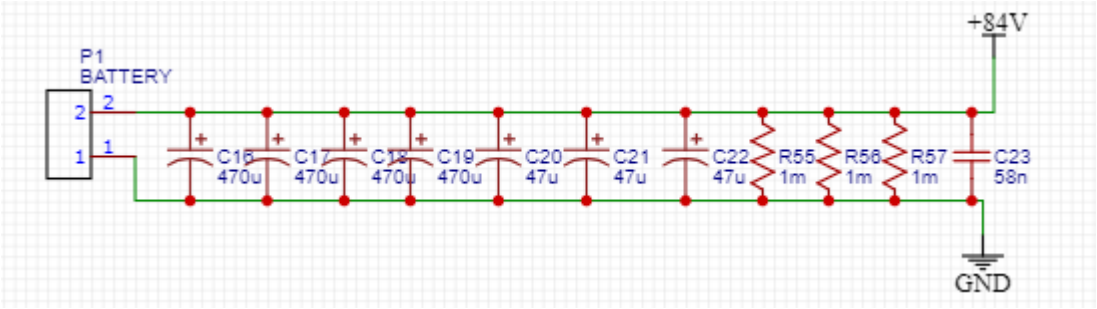



*Figure 6. DC BARA*

### 6.1.4 Communication

Incorporating Can-BUS into the circuit enables seamless communication with the vehicle control system, facilitating the exchange of essential data. Specifically, Hall sensor RPM data is transmitted to the vehicle control system via the Can-BUS interface. This integration allows for real-time transmission of crucial information regarding the rotational speed of the system, enabling the vehicle control system to make informed decisions and adjustments as needed. By leveraging Can-BUS communication, the circuit enhances its functionality and interoperability within the broader vehicle system, contributing to overall performance, efficiency, and control. The designed circuit is given below in Figure 7.

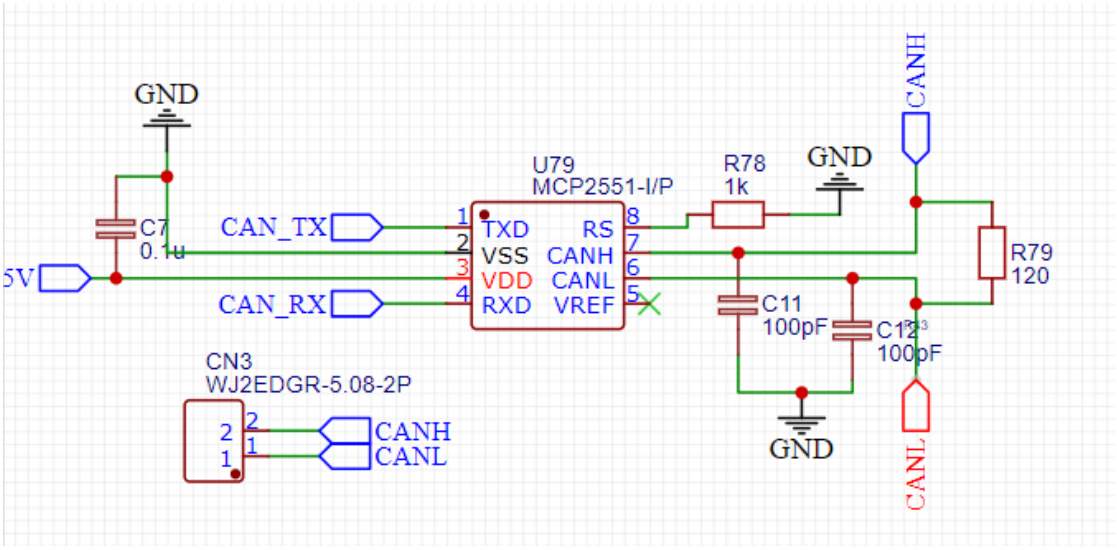



*Figure 7. Can-BUS Connection*

### 6.1.4 Microcontroller

The STM32 microcontroller, powered by 5 volts, serves as the central processing unit within the circuit. It plays a critical role in managing the engine's position information, reference speed, and direction data to trigger the appropriate MOSFETs effectively. Additionally, the STM32 offers versatility in communication options, allowing it to interact with external systems.

By leveraging its capabilities, the STM32 can transmit relevant information to the vehicle control system through multiple channels. Firstly, it can display data, such as RPM readings or system status, on the OLED screen integrated into the circuit, providing real-time feedback to users. Alternatively, the STM32 can utilize the Can-BUS interface to communicate directly with the vehicle control system, exchanging essential data for coordinated operation and control.

This multi-faceted approach empowers the STM32 to not only process crucial information for MOSFET triggering but also to facilitate seamless communication with external systems, enhancing overall system functionality, monitoring, and control within the vehicle environment. The designed circuit is given below in Figure 8.

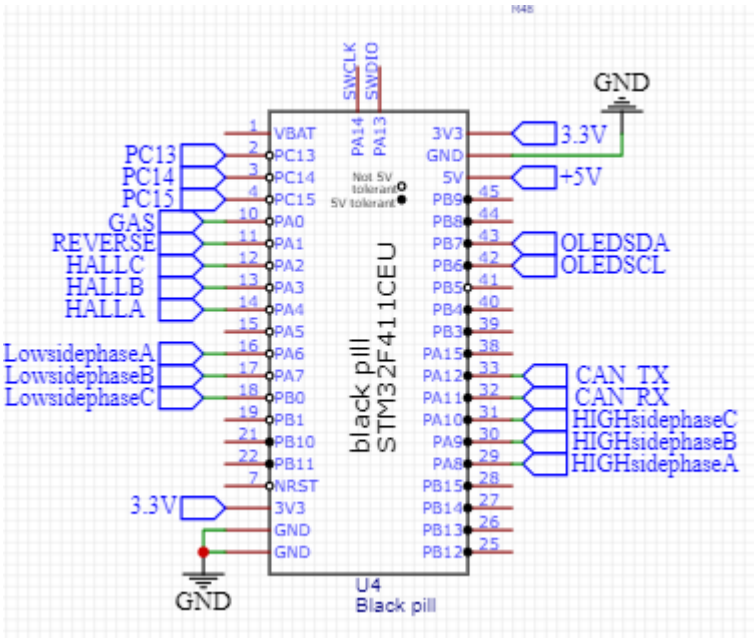



*Figure 8. STM32 Pins*

### 6.1.5 Microcontroller Connections

Within the circuit, ensuring compatibility between the 5V supply for the hall sensors and the STM32 microcontroller, which operates at 3.3 volts, is crucial. Therefore, a voltage level conversion mechanism is necessary to bridge this gap effectively. To accomplish this, the 74HC14 Schmitt trigger can be integrated into the circuit design. This component not only facilitates the conversion of the hall sensor output from 5 volts to 3.3 volts but also provides additional benefits in terms of signal conditioning and noise filtering.

Furthermore, during motor operation, the data obtained from the hall sensors may be susceptible to interference, potentially affecting the accuracy and reliability of the readings. To mitigate this issue, the Schmitt trigger functionality of the 74HC14 can help filter out unwanted noise and stabilize the sensor outputs, ensuring consistent and accurate data transmission to the STM32 microcontroller.

By incorporating the 74HC14 Schmitt trigger into the circuit, we not only address the voltage compatibility issue but also enhance the overall robustness and reliability of the system,

particularly in the presence of external interference during motor operation. This comprehensive approach ensures seamless integration and optimal performance of the hall sensor subsystem within the larger circuit architecture. The designed circuit is given below in Figure 9.

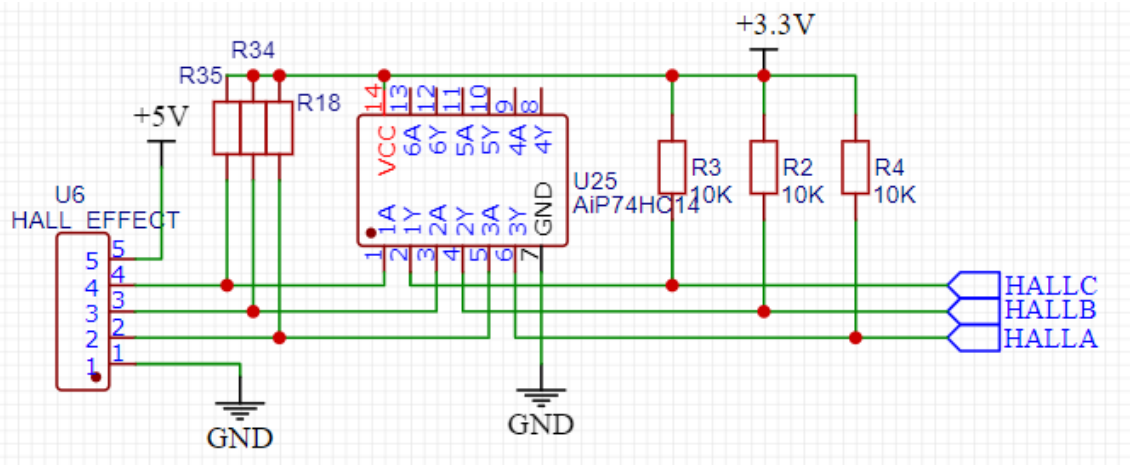



*Figure 9. 74HC14 Connection*

To accommodate accelerator pedals operating at varying voltage levels, several precautions have been implemented within the circuit design. Firstly, the maximum data voltage has been capped at 3.3 volts, ensuring compatibility with a wide range of accelerator pedal configurations. Additionally, a pull-down resistor has been incorporated to serve two main purposes: firstly, to prevent the system from operating when the accelerator pedal is not present, and secondly, to mitigate instability issues that may arise.

Furthermore, a low-pass filter has been integrated into the system to counteract high-frequency changes that could potentially affect the accuracy and stability of the accelerator pedal data. This filter effectively attenuates high-frequency components, ensuring that only the desired signal characteristics are passed through to the system, thereby enhancing reliability and performance. Precautionary measures not only ensure compatibility with different accelerator pedal voltages but also promote stability and reliability in the system's operation, safeguarding against potential instabilities and disturbances. Additionally, pulldown resistors were used as forward and reverse information. The designed circuit is given below in Figure 10.

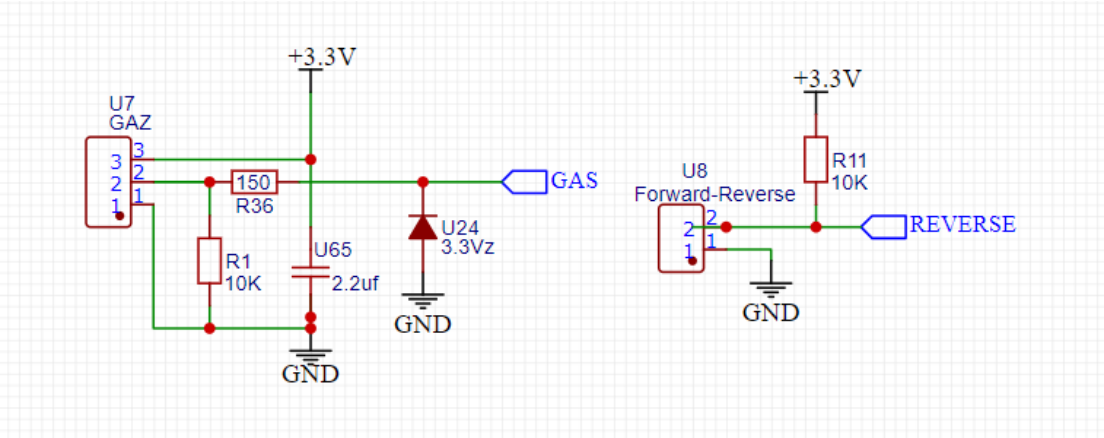



*Figure 10. Gas Pedal Connection*

Integration of LEDs into the circuit serves multiple purposes, increasing visibility and making it easier to monitor various system functions. Strategically positioned next to the MOSFETs, these LEDs provide visual indicators of the MOSFETs' status, allowing users to quickly assess their proper operation. By observing the LED illumination, users can determine whether the MOSFETs are operating as intended and assist in troubleshooting and maintenance. The designed circuit is given below in Figure 11.

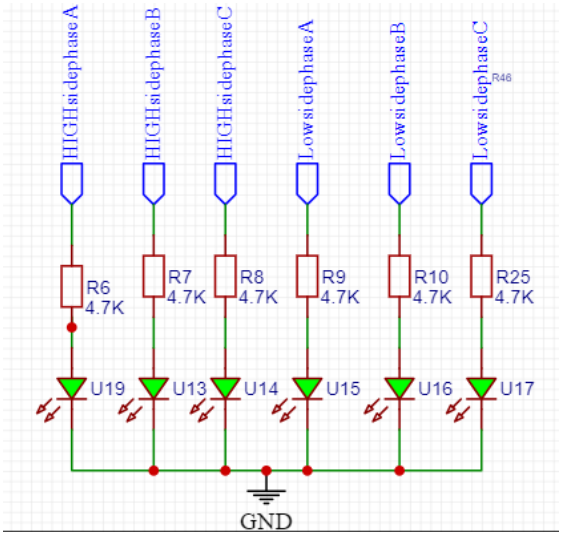



*Figure 11. Sides LEDs*

### 6.1.6 Regulators

To optimize the voltage supply, the battery system must operate in the range of 72V to 84V. Gate drivers for the motor driver (IR2127) require a supply range of 10V to 20V. A 3-stage reducer design will be applied to accommodate various interfaces within the motor driver. The step-down stages will be designed to convert from 84V to 15V, then to 5V, and finally to 3.3V. This design aims to minimize the heat and losses that occur during the conversion process from high voltage to low voltage. The reduced 15V output will power the gate drivers, 5V will feed the microprocessor and Hall sensors. 3.3 volts pull-up the signals entering the accelerator pedal and STM.

Capacitors are added to the input sections of regulator integrated circuits (ICs) to reduce noise and frequency aberrations caused by the power supply. Additionally, to give a linear voltage with less harmonic effects, capacitors are added to the output. It has been decided to employ the L7805, L7815, and LM1117 integrated circuits. Designs for portable heatsinks are used as a safety precaution against any heat-related problems. The designed circuits are given below in Figure 12,13 and 14.

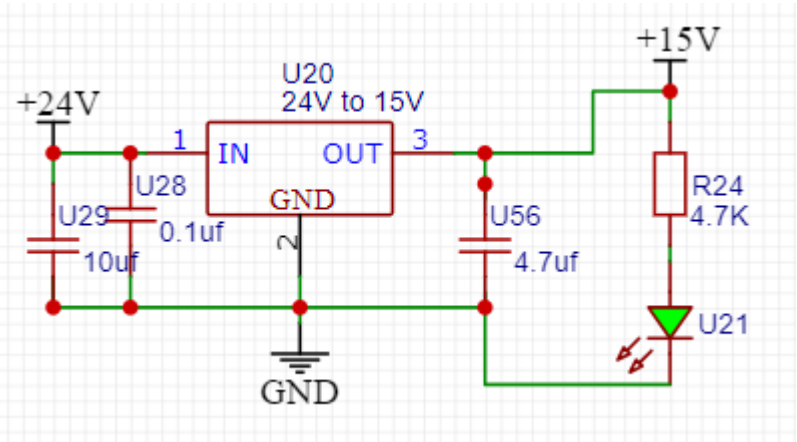



*Figure 12. 24V-15V Regulator*

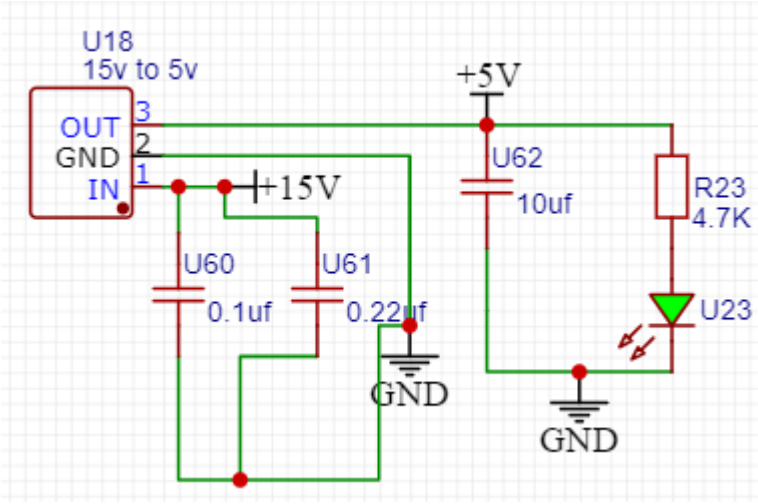



*Figure 13. 15V-5V Regulator*

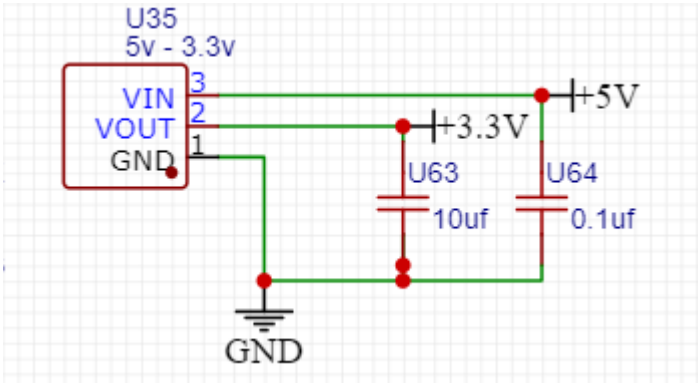



*Figure 14. 5V-3.3V Regulator*

### 6.1.6 PCB Design

Prior to finalizing the PCB design, thorough research was conducted, involving detailed examination of samples of BLDC motor drivers already in production. Drawing insights from this research, key decisions were made regarding the arrangement of components on the board to optimize performance and minimize potential interference. A primary consideration was to separate the control and power sections as much as possible to reduce interference between them. By positioning these sections at opposite ends of the board, the risk of electromagnetic interference (EMI) and noise coupling between components was mitigated, ensuring stable and reliable operation. to enhance grounding integrity and minimize ground loops, the ground connections of the microprocessor and the power section were consolidated at a single point

nearest to the power input. This centralized grounding approach helps to maintain consistent ground potential throughout the board, reducing the risk of signal distortion and voltage fluctuations.

To minimize losses in the MOSFETs and optimize switching performance, gate resistors were placed near the MOSFETs. This layout choice reduces parasitic inductance in the gate drive circuit, enabling faster and more efficient switching transitions. these strategic layout decisions based on extensive research and analysis contribute to the overall reliability, efficiency, and performance of the BLDC motor driver PCB design, ensuring optimal functionality and minimizing potential interference issues. The designed circuit is given below in Figure 15.

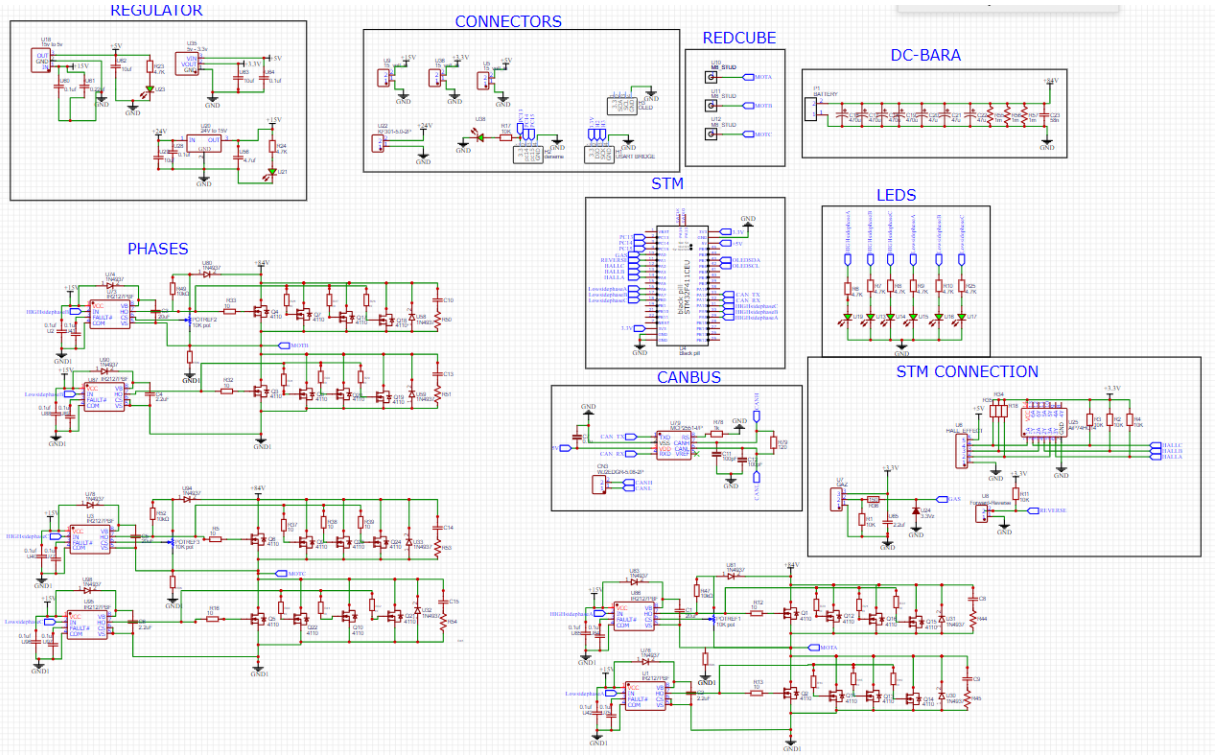



*Figure 15. Schematic Design*

The bottom layer of the PCB layout was assigned as the signal layer, and the top layer was assigned as the power layer. After the necessary calculations, the track widths were

determined as 1,000 mm for 5V lines, 1,500 mm for 15V lines, 15,000 mm for 84V power lines and 0.306 mm for signal lines. Angle designs were avoided during tracking routing to reduce EMI and ensure stable signal transmission.

To reduce noise and offer a reliable ground reference, the second layer of the PCB was dedicated to GND, and a plane design was used. To further minimize EMI, signal lines were carefully routed between GND planes and the GND of the microprocessor and power line were separated. Red cubes were used to transmit high amperes from the phases to the motor. Additionally, the enlarged GND plane helps reduce the heat dissipation of the board.

Metal copper was placed on the roads so that the power transmission line could carry high amperage, thus the durability of the roads was maximized by soldering copper to the road. The designed PCB is given below in Figure 16,17,18,19 and 20.

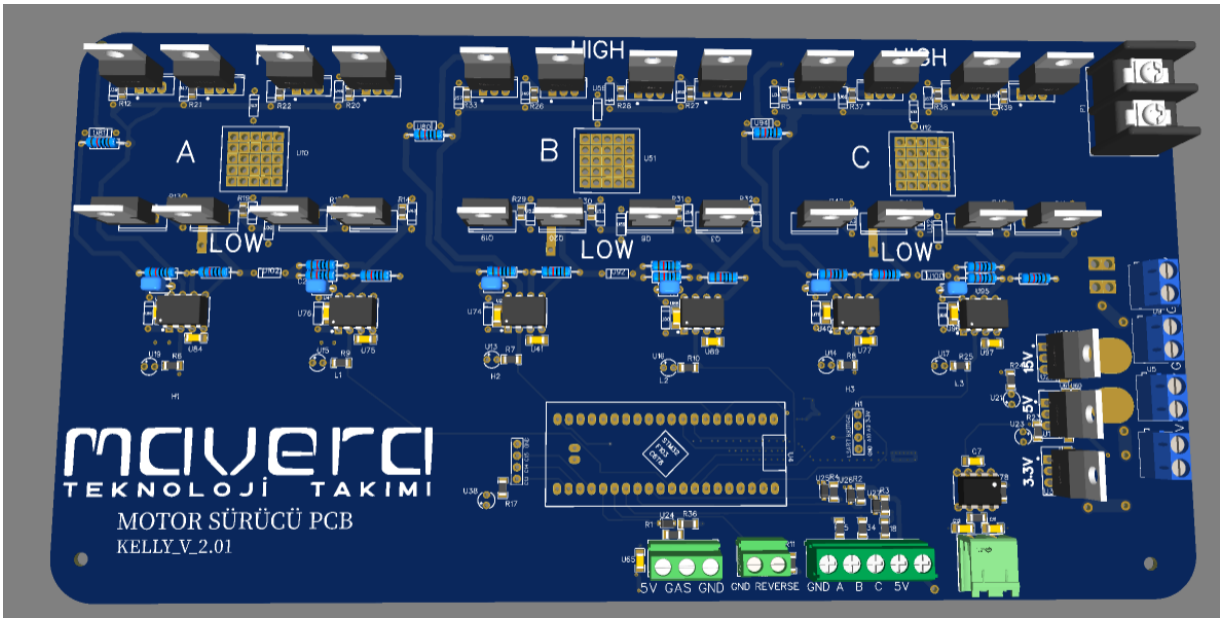



*Figure 16. 3D PCB View*

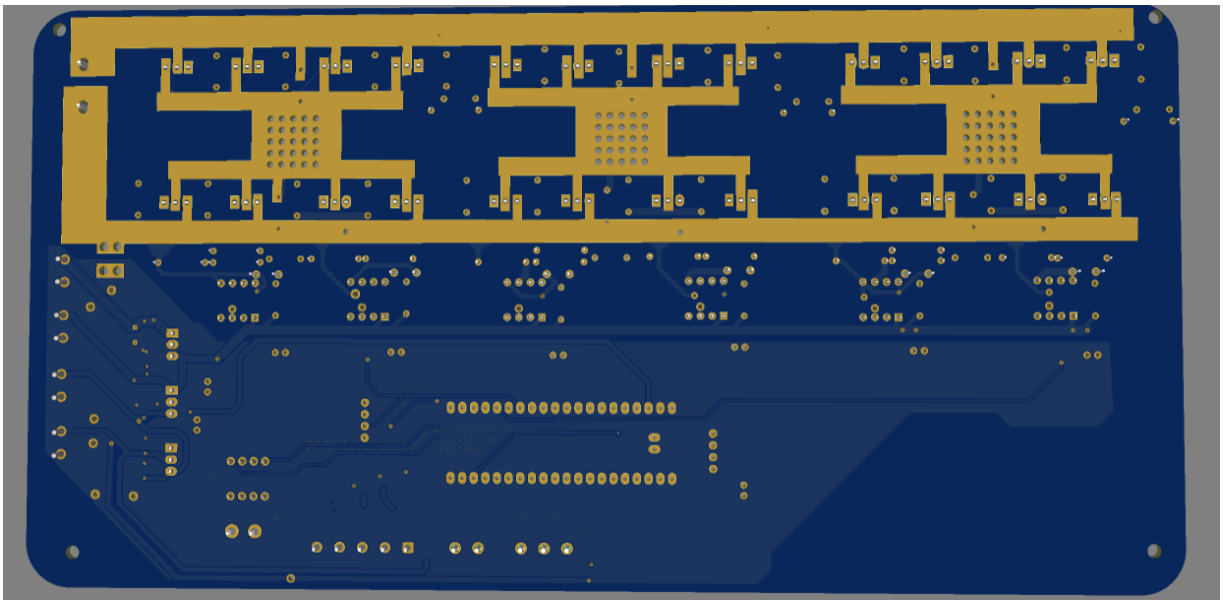


*Figure 17. 3D PCB View*

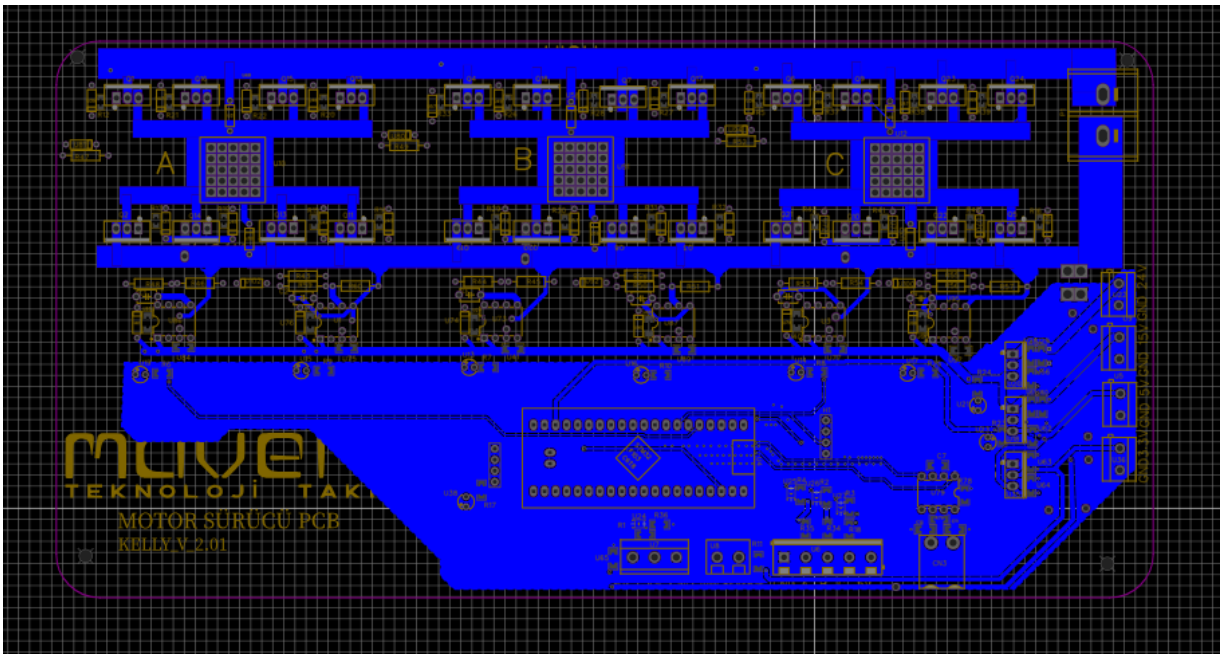



*Figure 18. Bottom Layer of PCB*

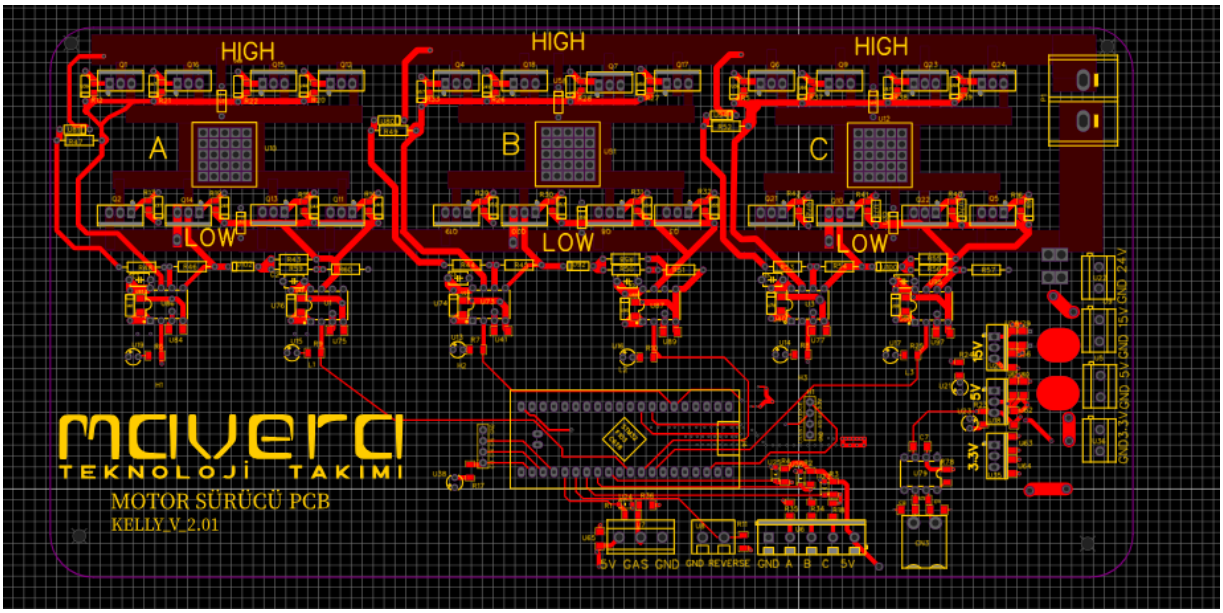



*Figure 19. Top Layer of PCB*

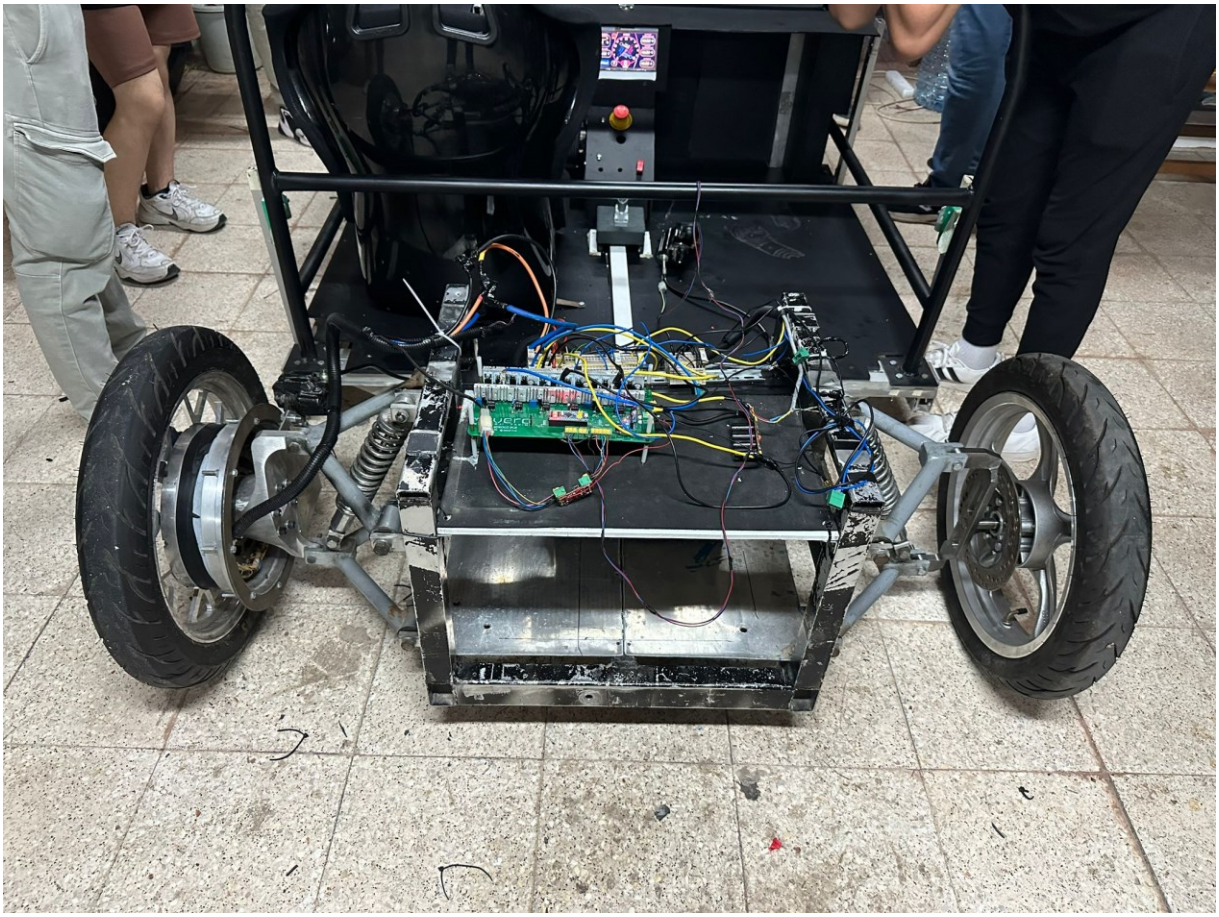


*Figure 20. The Result of PCB*

### 6.1.6 Cooling SYSTEM

While designing the cooling system, the primary reference point was the maximum temperature that the MOSFET could potentially reach. Extensive research and meticulous calculations indicated that the MOSFET could attain a temperature of approximately 300 degrees Celsius under certain conditions. To effectively manage and dissipate this heat, aluminium blocks were employed due to their excellent thermal conductivity properties. The cooling design aimed to significantly reduce the operational temperature of the MOSFETs, ensuring they would not exceed a target maximum temperature of 80 degrees Celsius. This targeted temperature reduction was critical not only for maintaining the reliability and longevity of the MOSFETs but also for ensuring the overall stability and efficiency of the entire system. By integrating these aluminium cooling blocks, the system can operate within safe thermal limits, thus preventing thermal-related failures and optimizing performance. Below is shown the thermal analysis of the temperature of mosfet. The cooling system is given below in Figure 21 and 22.

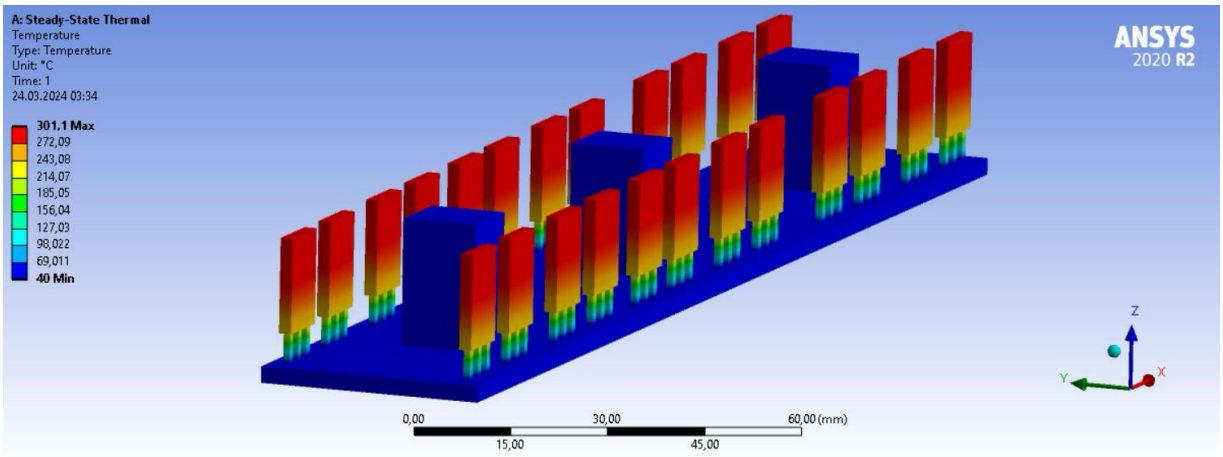



*Figure 21. Without Cooling System*

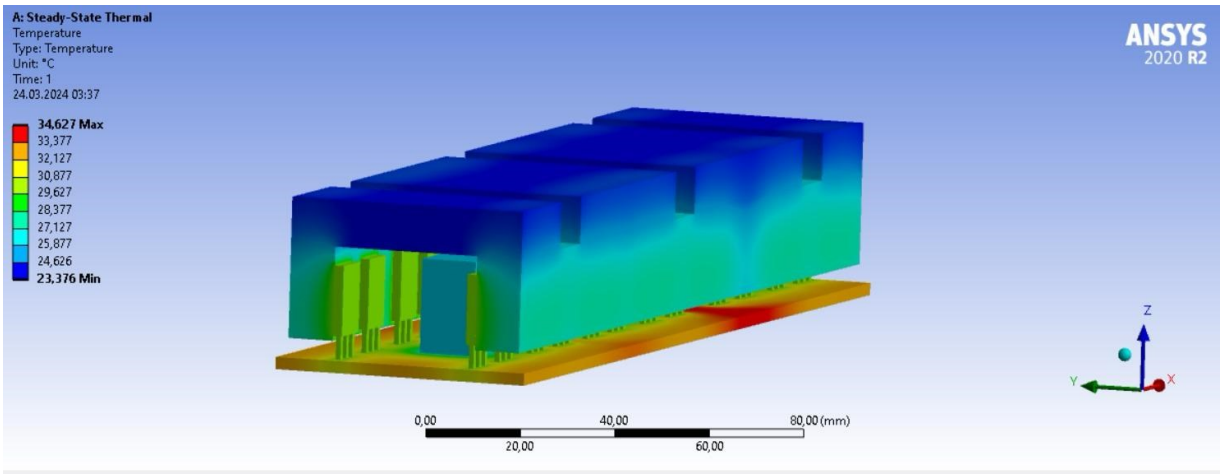



*Figure 22. With Cooling System*

As a result of the analysis, it was observed that the maximum temperature of the system at 0 degrees is 35 degrees. The test of cooling system is given below in Figure 21 and 22.

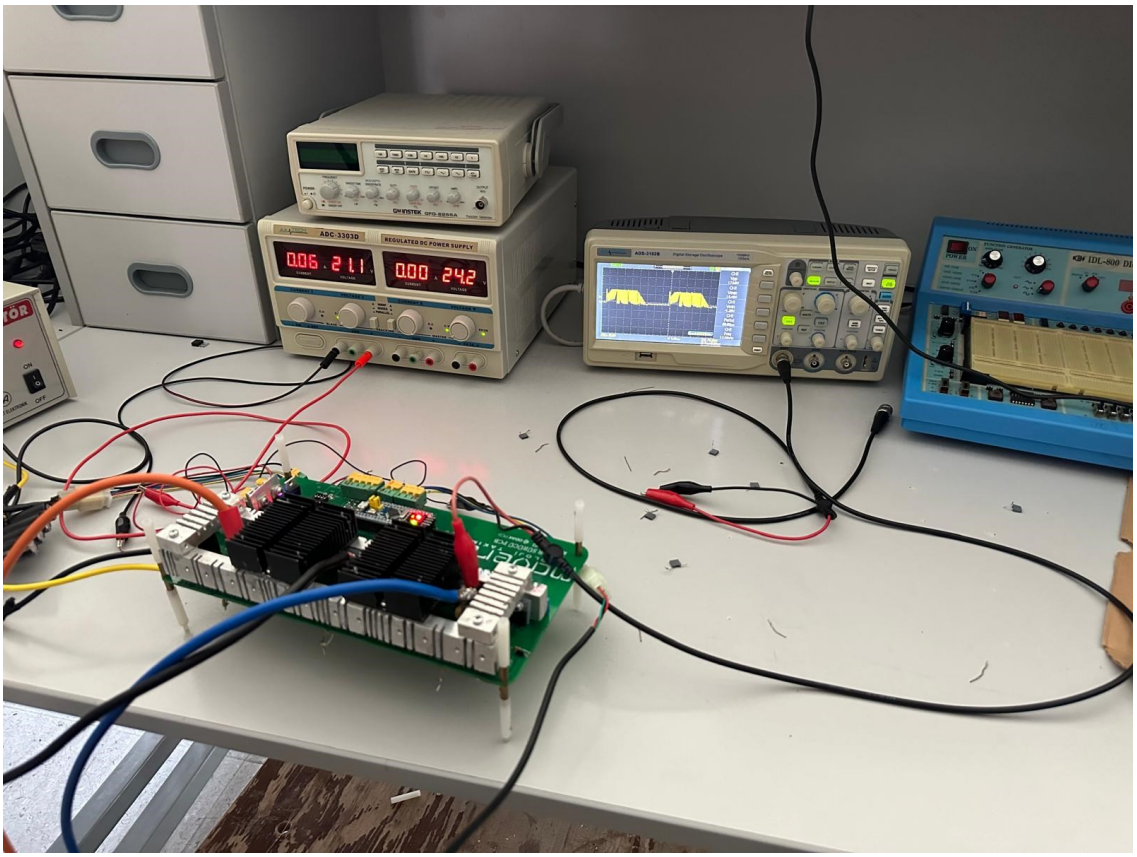


*Figure 23. Cooling System Testing*

To enhance the effectiveness of the cooling system, aluminium fins were incorporated to increase the surface area in contact with the air. This design modification allows for more efficient heat dissipation. By maximizing the exposed surface area, the aluminium fins facilitate better air circulation and heat exchange, thus enabling the cooling system to work more effectively. This improvement ensures that the MOSFETs maintain a safe operational temperature, contributing to the overall reliability and performance of the system.

## 6.2 Software

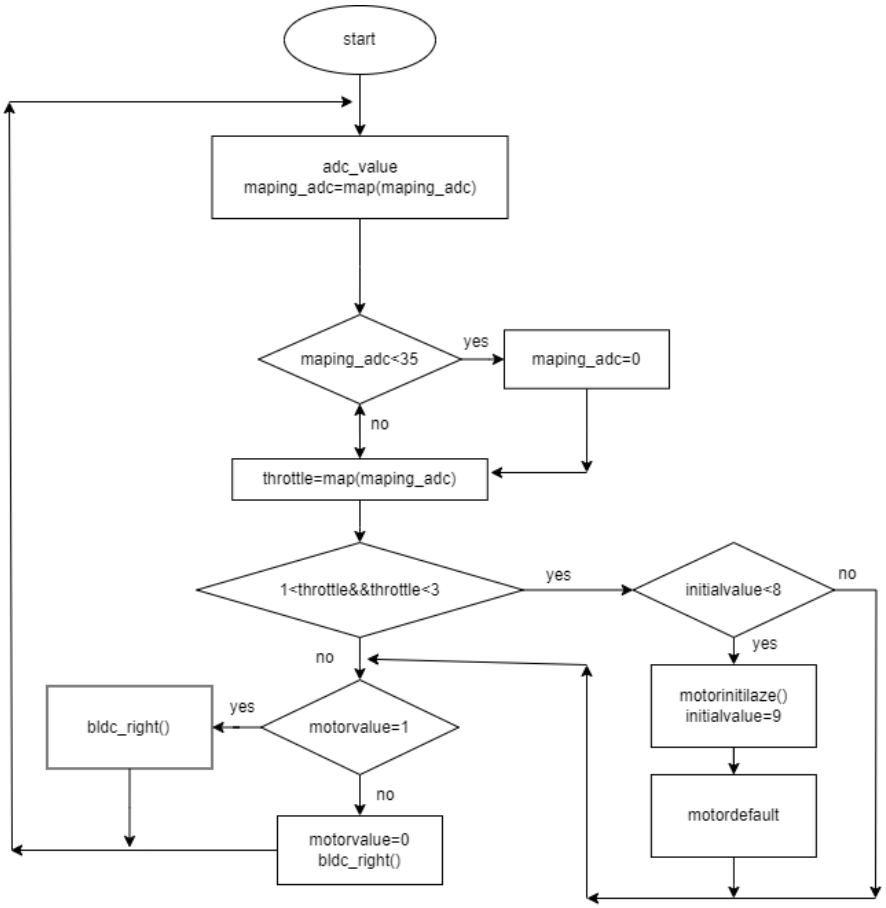



*Figure 24. Flowchart of BLDC Motor Driver*

Using the appropriate algorithms, the STM32F103C8 microcontroller was able to directly activate the phases while also detecting the PWM signal needed to operate the BLDC motor and its location. The motor's instantaneous location was determined by placing Hall effect sensors on it, and data was sent to the microcontroller. Using bit logic operations, the microcontroller processed the data from the Hall effect sensors to generate three-digit integers. In accordance with the signals from the Hall effect sensors positioned at a 120-degree angle to the motor, the motor coils were appropriately activated using the 3-digit numbers and the commutation table of the 6-step driving approach. The throttle lever has been added to the motor driver control card design to give motor speed control. The PWM signal's resolution was modified using the map function on the microcontroller in accordance with the PWM resolution ratio, utilizing analog data that was read from the potentiometer. There is a forward and backward button. The flowchart is given in Figure 24.

STM32cubeide program was used while coding. Pinout outputs are given below. To use the full speed of stm32, 72 MHz is also used.

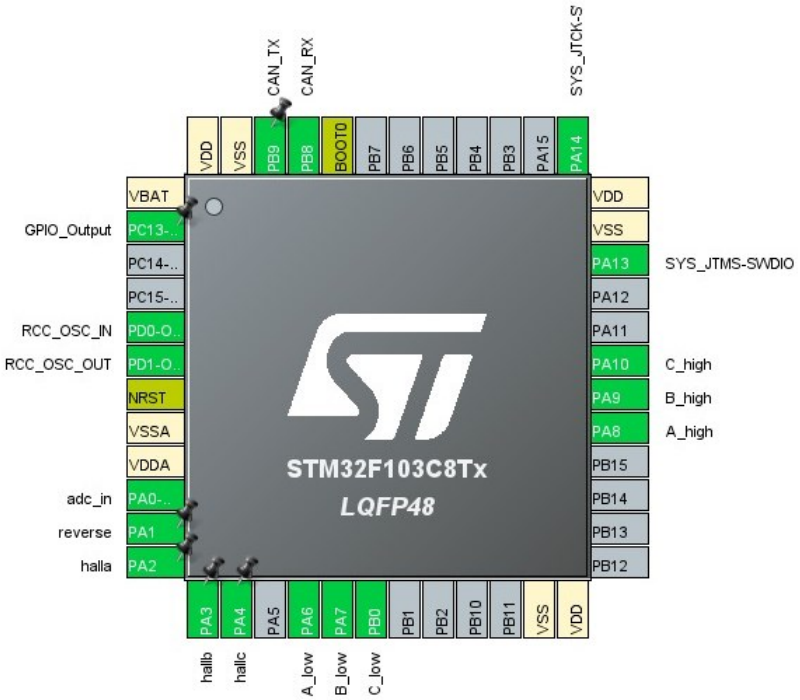



*Figure 25. STM32 Pinout Configuration*

In the system, data from the Hall sensors, ADC, and reverse pin are evaluated to generate outputs for the A, B, and C PWM channels. TIM1 and TIM3 are employed for PWM generation, playing critical roles in motor control. The Hall sensors provide real-time position feedback, while the ADC reads analog inputs such as the potentiometer for speed control. The reverse pin indicates the direction of rotation. The motor return code is executed within the TIM3 global interrupt, which ensures timely response and precise control of motor speed and position. Concurrently, the soft start functionality, which gradually ramps up the motor speed to prevent sudden jolts and reduce mechanical stress, is handled within the SysTick interrupt. This interrupt-based approach allows the system to manage different aspects of motor control efficiently, maintaining smooth and reliable operation.

Operating frequency is 16 kHz, TIM1 is configured as the master timer and TIM3 as the slave timer. TIM1 generates the primary PWM signals and controls the timing and synchronization of these outputs, while TIM3 operates in synchronization with TIM1 to handle specific motor control tasks, such as processing the motor return code. This configuration ensures precise timing and coordination between the master and slave timers, enabling effective motor control and smooth operation. The configuration is given Figure 26 and 27.

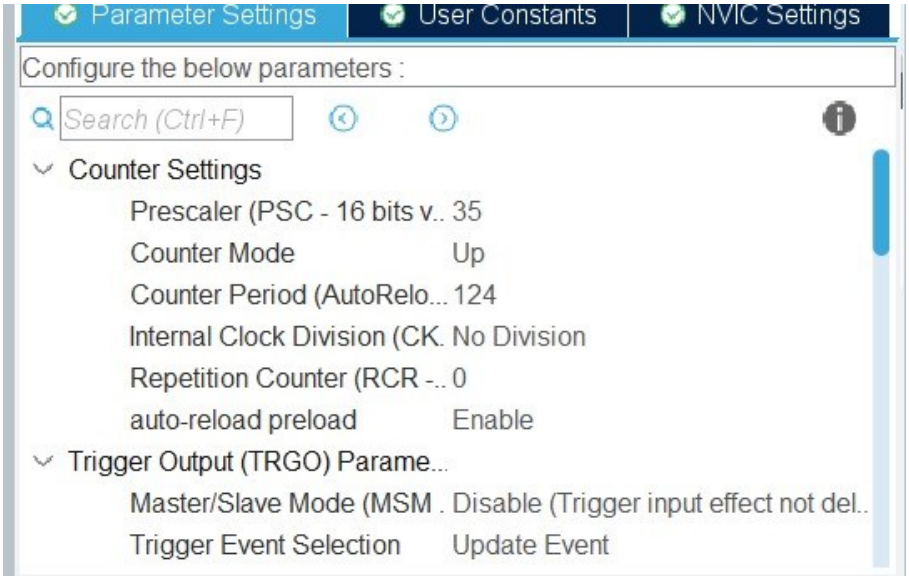



*Figure 26. Timer 1 Configuration*

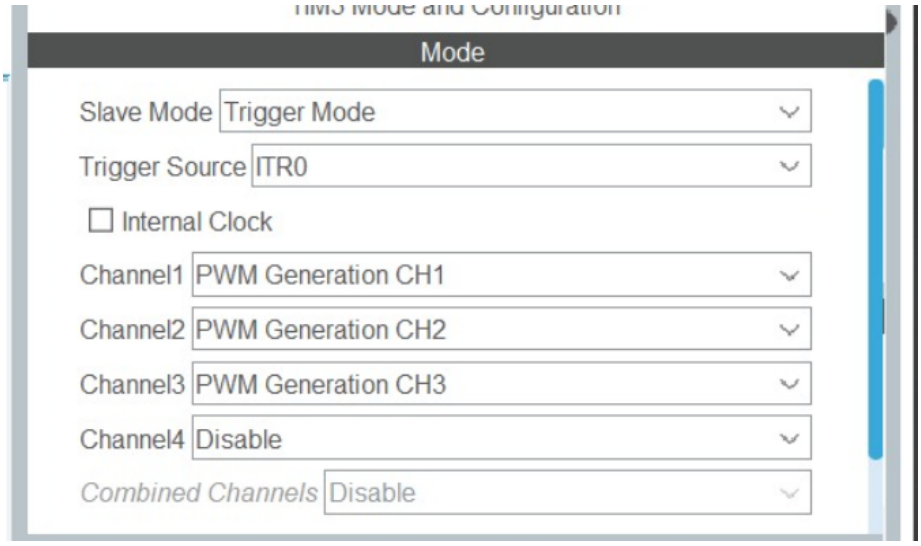



*Figure 27. Timer 3 Configuration*

The general codes of the motor driver are located in the motor and motor libraries. The functions used in the system are shown below in Figure 28.

```
2  #define __motor_H
3
4  #include "stm32f1xx_hal.h"
5  #ifdef __cplusplus
6  extern "C" {
7  #endif /* __cplusplus */
8  void BLDC_startcond(void);
9  void BLDC_phasecontrol(uint16_t adc);
10 void BLDC_motordefault(void);
11 void BLDC_HallSensorsGetPosition(void);
12 void BLDC_right(uint16_t adc);
13 void BLDC_left(uint16_t adc);
14 int BLDC_adcfilter(int deger);
15 int BLDC_adcfilter_value(void);
16 int BLDC_hallpos_value(void);
17 void BLDC_soft_start(int soft);
18 int BLDC_softstart_value(void);
19 void BLDC_speed(int deger);
20 void  BLDC_rpm_calc(int rpm);
21 int BLDC_rpm_calculated(void);
22 #ifdef __cplusplus
23 }
24 #endif /* __cplusplus */
25
```

*Figure 28. Motor Driver Functions*

General Functions

Motor Default: This function sets the outputs from the motor's Hall sensors to 0, effectively stopping the motor.

Soft Start: Initiates a gradual acceleration sequence to softly start the engine, reducing sudden strain and wear.

Right/Left Rotation: Changes the rotation direction of the motor.

The functions in the main code are given below in Figure 29 and 30.

```
449 void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim) {
450     if (htim->Instance == TIM3) {
451
452         filters_value = HAL_ADC_GetValue(&hadc1);
453         if (HAL_GPIO_ReadPin(GPIOA, reverse_Pin) == 1) {
454             BLDC_left(filters_value);
455         } else {
456             BLDC_right(filters_value);
457         }
458         //hall deÄŸerini okuma
459
460         //adc smooth deÄŸer okuma
461
462     }
463 }
464 }
465 void SysTick_Handler(void) {
466     HAL_IncTick();
467     //200ms
468
469     int sysyick_value = HAL_ADC_GetValue(&hadc1);
470     if (sysyick_value  <= 1200) {
471         BLDC_soft_start(0);
472     }
473     if (uwTick % 5 == 0) {
474         ++ctr;
475         BLDC_soft_start(ctr);
476         HAL_GPIO_TogglePin(GPIOC, GPIO_PIN_13);
477         ctr = BLDC_softstart_value();
478     }
479
480 }
```

*Figure 29. Main Codes*

```
    while (1) {
    /* USER CODE END WHILE */
        int sysyick_value = HAL_ADC_GetValue(&hadc1);
        if(sysyick_value>1170&&sysyick_value<1190){BLDC_startcond();}

    /* USER CODE BEGIN 3 */

    }
  /* USER CODE END 3 */
}
```

*Figure 30. Main Codes*

These codes aim to solve some problems that may occur in the proper starting and throttle of the motor.

The position information of the motor is taken as shown in the figure below. The driving method is applied in 6 steps and the appropriate MOSFETs are triggered according to this position information. This information is given in Figure 31.

**Table 3. Clockwise direction**

| Commutation vector | | | Vector | Hall sensor pattern definition | | | Hall sensor pattern result |
|---|---|---|---|---|---|---|---|
| Phase A | Phase B | Phase C | | Hall Sensor C | Hall Sensor B | Hall Sensor A | |
| NC | $+V_{DCB}$ | $-V_{DCB}$ | A | 1 | 0 | 1 | 5 |
| $-V_{DCB}$ | $+V_{DCB}$ | NC | B | 1 | 0 | 0 | 4 |
| $-V_{DCB}$ | NC | $+V_{DCB}$ | C | 1 | 1 | 0 | 6 |
| NC | $-V_{DCB}$ | $+V_{DCB}$ | D | 0 | 1 | 0 | 3 |
| $+V_{DCB}$ | $-V_{DCB}$ | NC | E | 0 | 1 | 1 | 2 |
| $+V_{DCB}$ | NC | $-V_{DCB}$ | F | 0 | 0 | 1 | 1 |

**Table 4. Counterclockwise direction**

| Commutation vector | | | Vector | Hall sensor pattern definition | | | Hall sensor pattern result |
|---|---|---|---|---|---|---|---|
| Phase A | Phase B | Phase C | | Hall Sensor C | Hall Sensor B | Hall Sensor A | |
| NC | $-V_{DCB}$ | $+V_{DCB}$ | D | 1 | 0 | 1 | 5 |
| $-V_{DCB}$ | NC | $+V_{DCB}$ | C | 0 | 0 | 1 | 1 |
| $-V_{DCB}$ | $+V_{DCB}$ | NC | B | 0 | 1 | 1 | 3 |
| NC | $+V_{DCB}$ | $-V_{DCB}$ | A | 0 | 1 | 0 | 2 |
| $+V_{DCB}$ | NC | $-V_{DCB}$ | F | 1 | 1 | 0 | 6 |
| $+V_{DCB}$ | $-V_{DCB}$ | NC | E | 1 | 0 | 0 | 4 |

*Figure 31. 6 Steps*

Independent Unipolar PWM Modulation" technique is distinguished by its ease of application and control. Compared to other modulation techniques, it subjects the MOSFET to the least amount of stress, making it highly efficient and minimizing losses. Its user-friendly nature makes it a preferred choice in various applications, providing flexibility in system design. One notable feature of this technique is the absence of active braking, which simplifies the overall system architecture.

In Independent Unipolar PWM Modulation, the switching events are arranged in such a way that only one switch is active at a time, reducing the switching losses and thermal stress on the components. This leads to improved reliability and longevity of the system. The technique is especially advantageous for low-frequency applications, where it ensures smooth and precise control of the output voltage and current.

Furthermore, the simplicity of this modulation technique allows for easier implementation in digital control systems, making it suitable for modern microcontroller and DSP-based designs. It also facilitates straightforward integration with other control algorithms and enhances the overall efficiency of power electronic converters.

Overall, Independent Unipolar PWM Modulation offers a robust and efficient solution for various power conversion needs, balancing performance, simplicity, and reliability. Its inherent advantages make it a valuable choice for engineers seeking to optimize their power electronics designs. The modulation is given Figure 32.

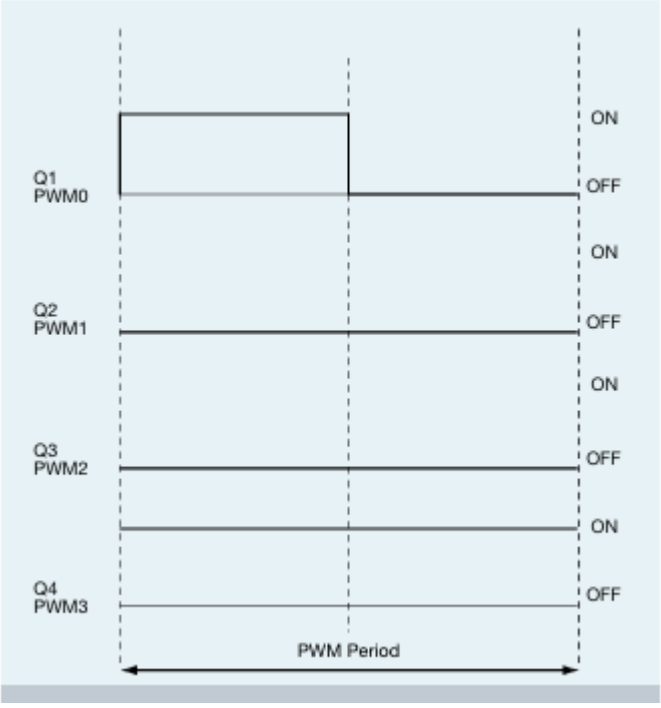



*Figure 32. Independent Unipolar PWM Modulation*

### 6.3 Tools

- PSIM
- Multisim
- STM32CubeIDE
- Proteus
- Ansys

## 7. EXPERIMENTS

### 7.1 Simulation Results

Before designing the motor driver circuit control card, extensive research was conducted on PWM modulation issues, and simulations were carried out using LT-Spice. These preliminary investigations and simulations helped in understanding the intricacies of PWM modulation and its impact on the system's performance. Following this, the circuit was established using PSIM to verify its functionality and performance in a simulated environment. Once the design was validated, the circuit was then printed according to the PCB layout specifications, ensuring that all components were correctly placed and connected to achieve optimal performance. This meticulous approach ensured that the final motor driver circuit control card was both efficient and reliable. This working is given Figure 33 and 34.

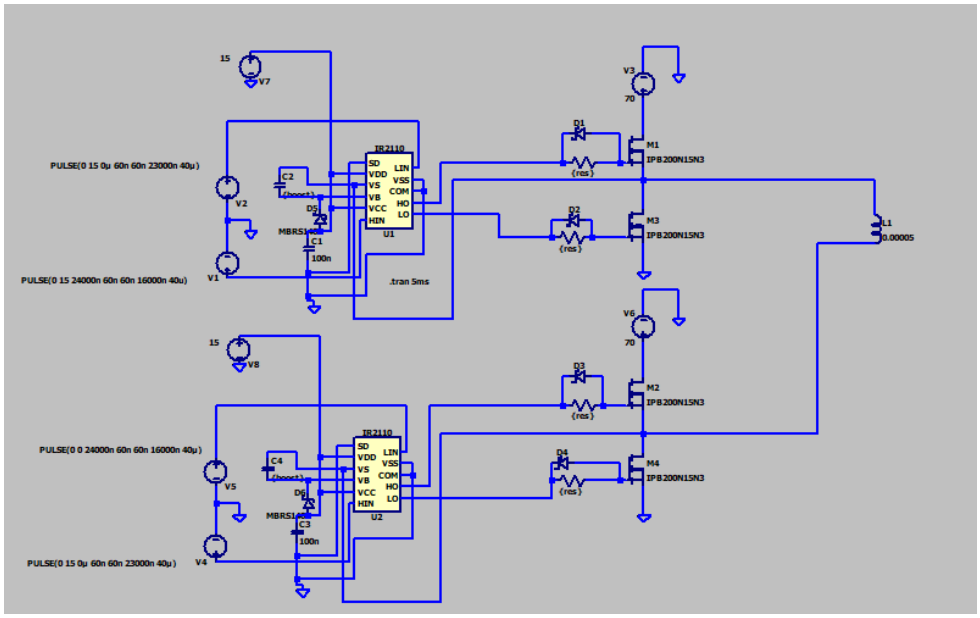



*Figure 33. H-bridge*

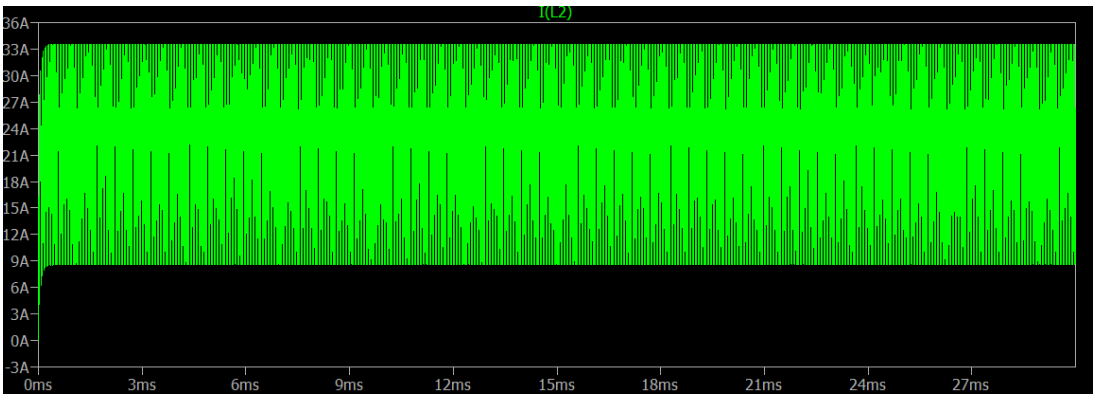



*Figure 34. H-Bridge Motor Output*

## 7.2 PCB Testing

First the SMD components of the PCB are soldered. Circuit is given Figure 35.

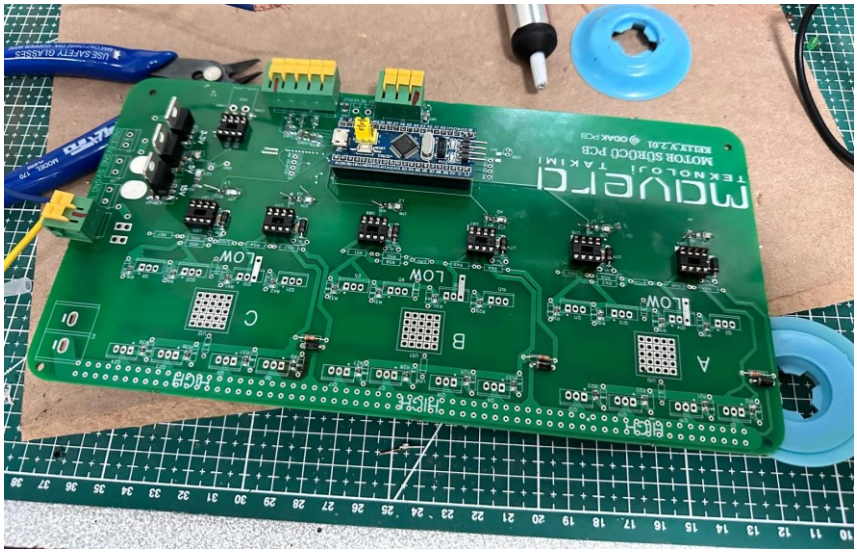


*Figure 35. BLDC Motor Driver PCB*

Afterwards, PWM outputs were checked with an oscilloscope. The testing is given Figure 36.

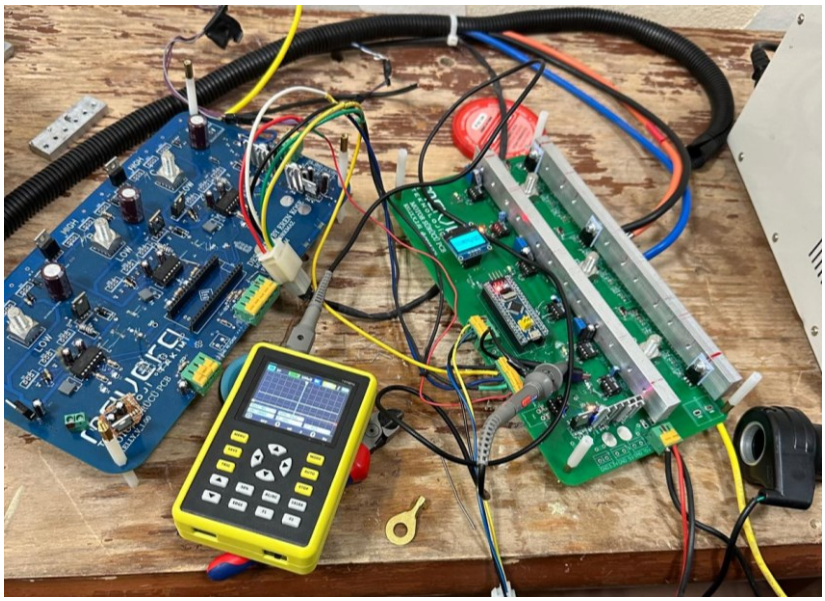


*Figure 36. PWM Controlling*

Once the PCB was completed, testing was done with the engine. The testing is given Figure 37.

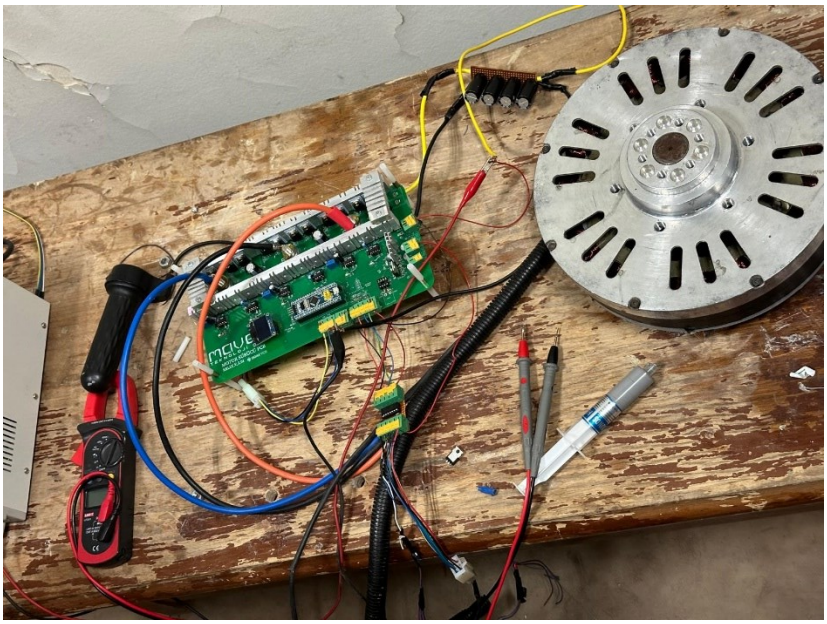


*Figure 37. MOTOR Testing*

As a result of the tests carried out with the engine, the snubber circuit was calculated. The testing is given Figure 38,39 and 40.

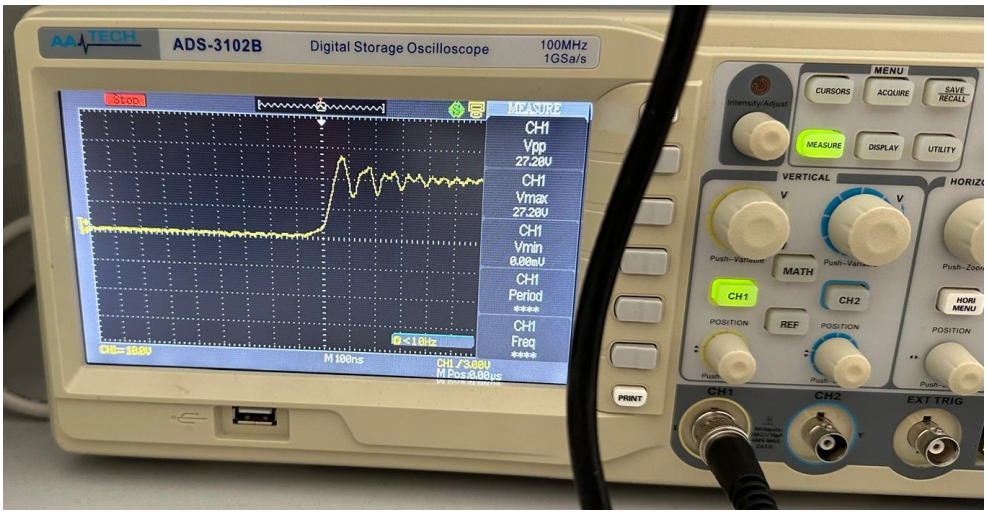



*Figure 38. Without Snubber*

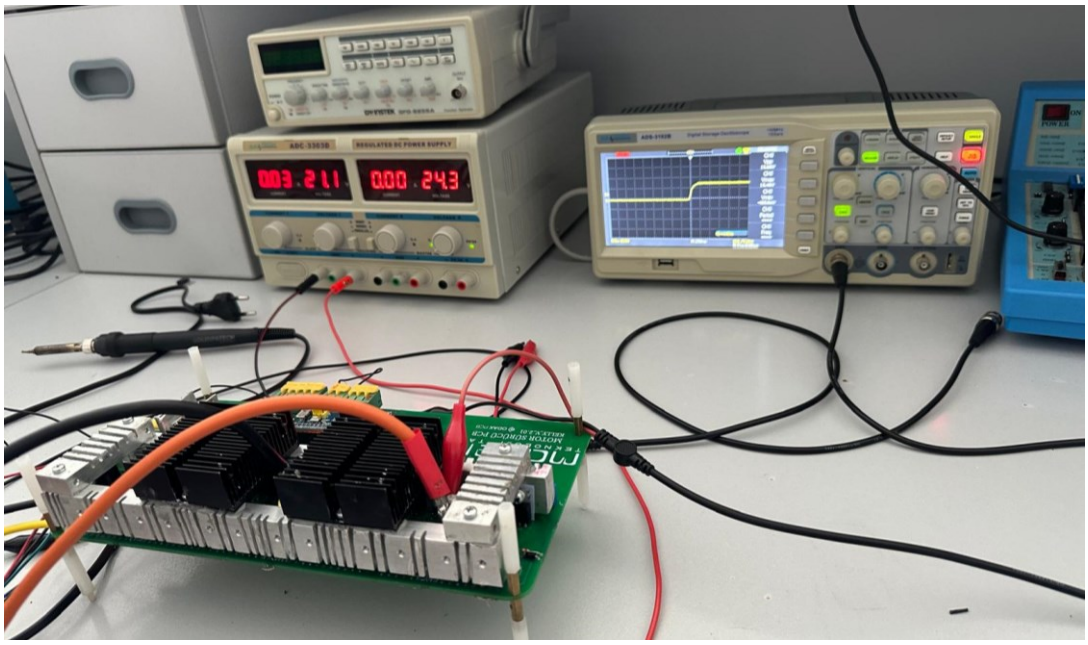


*Figure 39. PCB With Snubber*

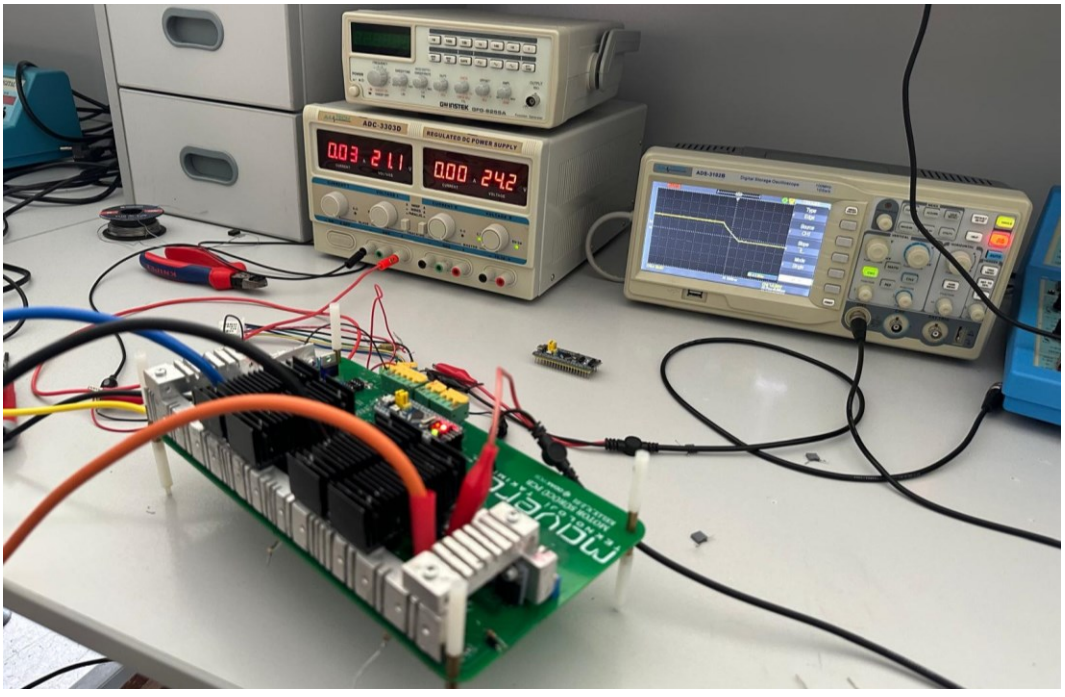


*Figure 40. PCB With Snubber*

As a result of the snubber test and other tests, tests were carried out on the vehicle. The result of working is given Figure 40 and 41.

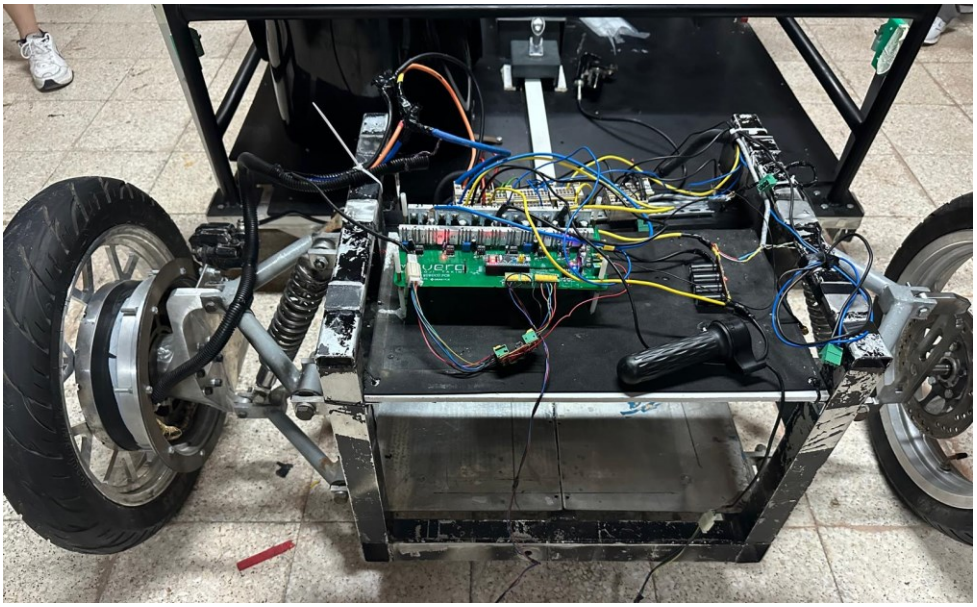


*Figure 41. BLDC Motor Driver Car Testing*



*Figure 42. Result of Testing*

# 8.PROJECT PLAN

- **Work Package 1 – Literature Search:** This is the starting section of the project. Every team member performed a review of the literature. The elements that were to be utilized and the research that was done were looked at. This section lasted 6 weeks.

- **Work Package 2 – Simulation:** In this section, the circuits taken from the literature review were simulated and the integrated circuits that should be included in the hardware were decided based on the outputs, and improvements were made by comparing the resulting outputs with other circuits. This section lasted 7 weeks.

- **Work Package 3 – Hardware Design:** In this section, PCB design and schematic drawing are made. PCB designed via easyEDA. This section lasted 2 weeks.

- **Work Package 4 – Software Design:** In this section, software has been created for the STM32F103C8T6 card. The program in which the software is written is STM32CUBE. This section lasted 10 weeks.

- **Work Package 5 – PCB Assembly Process:** In this section, the PCB was typed, and it took about 1 week.

- **Work Package 8 – Test on BLDC Motor:** In this section, integrated tests of the software written, and the hardware created are carried out. This section lasted 12 weeks.

*Table 3. Resource Assignments for Work Packages*

| Work Package | Resource | Duration (weeks) |
|---|---|---|
| 1 | All member | 6 |
| 2 | All member | 7 |
| 3 | Mustafa,Erdi,Eren | 2 |
| 4 | Eren,Ersin,Mustafa | 10 |
| 5 | Erdi | 1 |
| 6 | All member | 12 |
| ***PROJECT COMPLETITION TIME*** | | 38 |

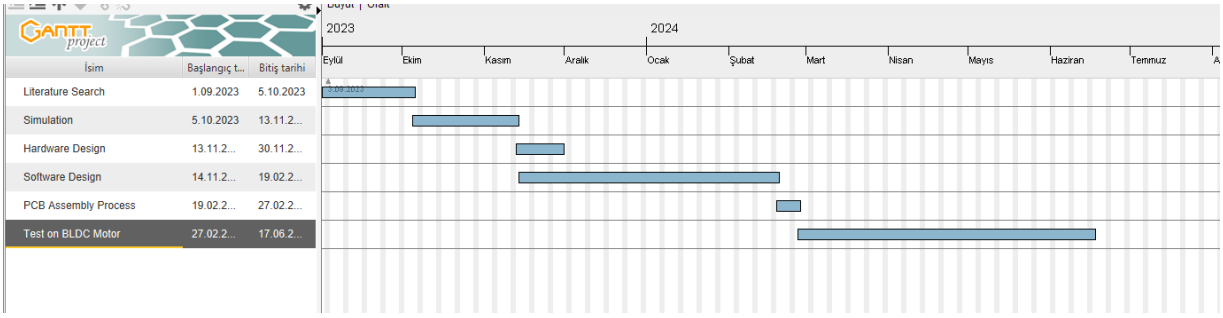



*Figure 43. Gantt Diagram of The Project*

# 9. CONCLUSION

The system has been meticulously developed to drive a 3.5 kW BLDC electric motor, utilizing a circuit board that is engineered to achieve high efficiency and high-speed performance. Throughout the development process, extensive analysis of the switching sequences allowed us to identify and resolve several critical issues. These included dead time, cross-conduction, MOSFET overheating, and malfunctioning hall effect sensors. By addressing these issues and incorporating the fixes into the system, we ensured reliable and efficient operation.

During the PCB design phase, the substrate was entirely coated with copper to serve as a ground plane, significantly enhancing the grounding of the circuit. Additionally, vias were strategically added to facilitate easier access to the ground line, further improving the overall circuit stability and performance.

The system is equipped with comparator opamps integrated into the gate driver low-side outputs (IR2127), which enable precise current sensing. To protect the system from reverse voltage currents generated by the motor, high-current flyback diodes were installed in parallel with the MOSFETs. Moreover, a snubber circuit was added to protect the circuit from high voltage ripples, ensuring the longevity and reliability of the components.

Understanding the initial conditions of the system, 1M resistors were added in parallel with the MOSFETs to address the empty state of the bootstrap capacitors when the motor first

starts up. This design choice ensures that the capacitors are appropriately charged, allowing for smooth initial operation.

To further enhance the system's performance, a cooling system was designed to prevent the negative effects of excessive stress on the MOSFETs, thereby avoiding overheating and potential failures. Additionally, a Schmitt trigger was incorporated into the system to improve the accuracy of data received from the hall sensors. This enhancement ensures that the signals are processed more accurately, contributing to the overall precision and reliability of the motor control.

Currently, the system includes all necessary modules to ensure robust operation. The insights and knowledge gained from this design will be utilized to create a more compact and efficient version in future iterations. However, even in its current state, the system is fully capable of driving the vehicle, meeting all performance and safety standards. This comprehensive approach to design and integration underscores commitment to delivering a reliable and high-performing motor driver system.

## 10. REFERENCES

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